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**ΟΙΚΟΝΟΜΙΚΟ
ΠΑΝΕΠΙΣΤΗΜΙΟ
ΑΘΗΝΩΝ**



**ATHENS UNIVERSITY
OF ECONOMICS
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Master's Program: Data Science
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MCO vs MIA A Tale of Two Airports

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Introduction

This report presents a comparative data story for two of the busiest airports in the United States—Orlando International Airport (MCO) and Miami International Airport (MIA)—using detailed, flight-level operational data from the Harvard Dataverse U.S. commercial flights dataset covering the period 2004–2008.

The primary objective of this analysis is to assess the operational status and progress of these two airports over time, highlight their respective strengths and weaknesses, and propose data-driven recommendations for performance improvement. Given their geographic proximity and shared role as major hubs in Florida, MCO and MIA provide an informative basis for comparative analysis, particularly in terms of demand seasonality, network structure, and delay propagation.

Airports operate as complex systems influenced by multiple interdependent factors, including airline scheduling strategies, infrastructure capacity, weather conditions, air traffic control constraints, and broader network effects. In this study, we focus on performance indicators that directly affect both passengers and operators, such as on-time arrival performance, delay duration and severity, cancellation rates, and the underlying causes of delays. Temporal patterns—by year, season, and time of day—are examined to better understand congestion and resilience dynamics.

The analysis is based on a cleaned and processed subset of the original dataset, consisting of approximately 1.6 million flights involving MCO and MIA. It should be noted that data for 2008 only cover the first four months of the year; therefore, results involving 2008 are interpreted with appropriate caution.

To communicate our findings effectively, we employ a visual storytelling approach supported by more than twelve visualizations, including time-series plots, distributional analyses, heatmaps, route maps, and interactive dashboards. These visualizations are produced using both code-based tools (Python) and interactive software (Tableau). Together, they support a balanced narrative that highlights both positive performance trends and operational challenges, ultimately guiding a set of targeted and actionable recommendations for each airport.

The following report is organized in section as follows:

- **Section 1** presents the main **Exploratory Data Analysis (EDA)**, providing an overview of the dataset and a comparative assessment of Orlando International Airport (MCO) and Miami International Airport (MIA). This section highlights key performance differences and identifies areas that warrant deeper investigation.
- **Section 2** conducts a **focused analysis of flight delays**, examining temporal and operational patterns, comparing delay behavior across the two airports, and discussing potential explanations and improvement strategies.
- **Section 3** broadens the scope of the analysis to **flight cancellations**, with particular emphasis on the role of **meteorological factors**, offering insights into how weather-related disruptions affect airport reliability.
- Finally, **Section 4** presents an integrated **interactive dashboard** that synthesizes the main findings of the study while enabling further exploratory analysis. This dashboard allows

users to dynamically examine airport performance across multiple dimensions and to uncover additional insights beyond those discussed in the preceding sections.

1. Exploratory Data Analysis: MCO vs MIA Operational Overview

This section presents an exploratory data analysis (EDA) of Orlando International Airport (MCO) and Miami International Airport (MIA), examining fundamental operational patterns and characteristics during 2004-2008. Using U.S. Department of Transportation flight data, we analyze traffic volumes, route networks, and temporal patterns to establish a baseline understanding of how these two major Florida airports operate. The EDA focuses on three key dimensions: overall traffic trends, destination connectivity, and daily operational rhythms. These foundational insights reveal the distinct roles each airport plays in the U.S. aviation network—MCO as a high-volume leisure destination and MIA as a strategically positioned international gateway.

Through three core visualizations, we examine monthly departure patterns to understand traffic volumes and seasonality, route networks to identify primary connections and market focus, and hourly/daily departure distributions to reveal operational scheduling patterns. This analysis provides the essential context for understanding more complex operational challenges such as delay patterns, performance metrics, and external disruptions covered in subsequent sections.

1.1. Monthly Departures - MCO vs MIA

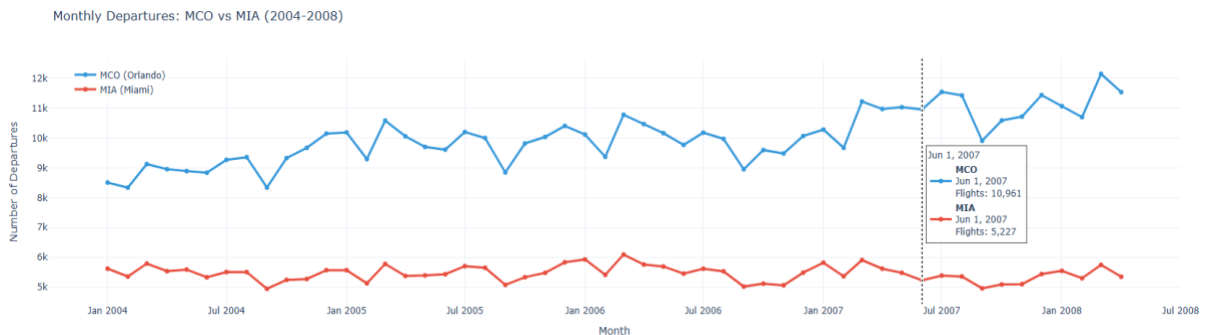


Figure 1.1: Comparison of Monthly Departures MCO vs MIA (2004-2008).

This visualization tracks monthly departure volumes from both airports over the entire study period to identify traffic patterns, growth trends, and the impact of incomplete 2008 data. Understanding monthly patterns helps airports plan resources and identify peak operational periods.

Key Findings:

- MCO consistently handles ~2x the departure volume of MIA throughout the entire period (averaging 10,000 vs 5,000 monthly departures).
- Clear seasonal patterns emerge: Both airports show summer peaks (June-August), indicating Florida's role as a vacation destination.
- Steady growth from 2004-2007: Both airports experienced consistent year-over-year increases until 2008.
- 2008 data is incomplete (January-April only), showing the expected volume drop.

- MCO's volume advantage remains stable across all seasons and years.

The 2:1 volume ratio between MCO and MIA reflects Orlando's position as a pure leisure/tourism destination versus Miami's more mixed market. MCO serves the massive Central Florida theme park industry (Disney World, Universal Studios), generating consistent high volumes. The synchronized seasonal peaks at both airports demonstrate Florida's overall tourism cycle, but MCO's higher absolute numbers show the stronger draw of Orlando's attractions. The growth trajectory ending in 2008 coincides with the beginning of the financial crisis, which would significantly impact leisure travel in subsequent years.

1.2. Top 10 Destinations from MCO vs MIA

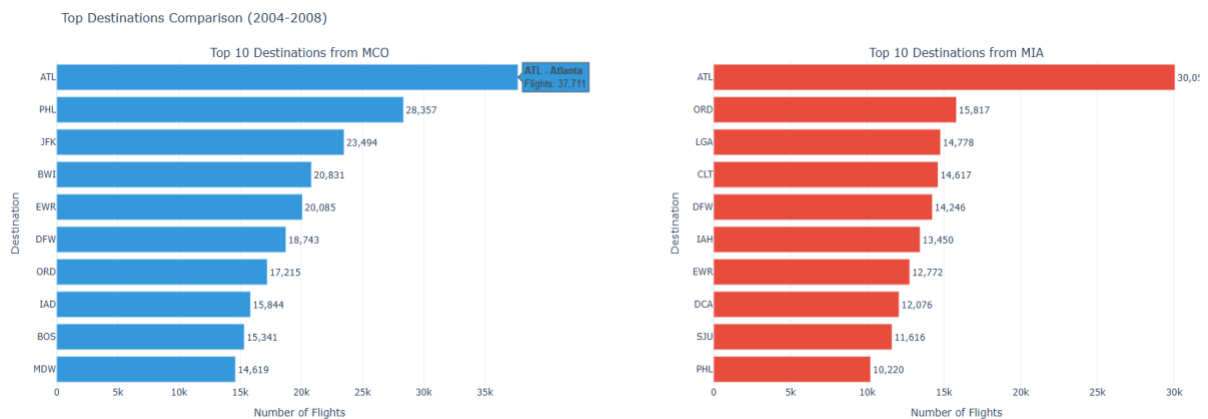


Figure 1.2: Comparison of Top 10 Destinations from MCO & MIA (2004-2008).

Route network analysis reveals the fundamental differences in how these airports function within the U.S. aviation system. By examining the top destinations, we understand whether an airport serves primarily as a local origin/destination point or as a connecting hub.

Key Findings:

- MCO focuses on major domestic hubs: ATL (Atlanta), ORD (Chicago), DFW (Dallas) dominate, indicating strong connecting traffic.
- MIA shows concentrated Northeast corridor traffic: JFK, LGA, and EWR (all New York area) feature prominently.
- ATL is #1 for both airports but with vastly different volumes (71,000 flights from MCO vs 30,000 from MIA).
- MCO serves a broader geographic spread: Destinations span the entire continental U.S.
- MIA's top routes reflect international gateway function: Strong Northeast connections support Caribbean/Latin American connections.

The destination patterns reveal two different business models. MCO operates as a major domestic tourist destination with strong links to connecting hubs nationwide—visitors from all over the U.S. fly through major hubs to reach Orlando's attractions. MIA's concentration on Northeast airports

reflects its role as the primary U.S. gateway to Latin America and the Caribbean; passengers from New York connect through Miami to reach international destinations. The Atlanta dominance at both airports shows Delta's hub power, but MCO's higher volume reflects Orlando's pull as a final destination rather than a connection point. These fundamentally different network structures require different operational strategies and airline partnerships.

1.3. Departure Time Heatmap (By Day and Hour)

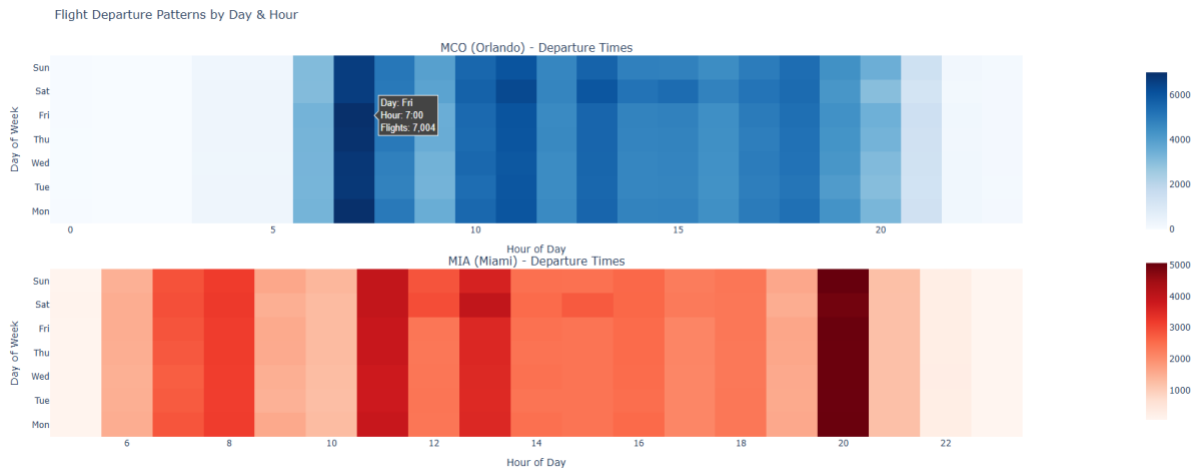


Figure 1.3: Comparison of Departure Time Heatmaps (by Day & Hour) for MCO & MIA (2004-2008).

This heatmap visualization reveals when flights actually depart throughout the week, exposing daily operational rhythms and showing how airports manage capacity across different days and times. Peak hour identification is critical for resource allocation and congestion management.

Key Findings:

- Early morning rush (5-7 AM) is the busiest period at both airports—airlines maximize aircraft utilization with dawn departures.
- Evening peak (5-8 PM) creates a second daily surge, particularly at MCO.
- Weekend patterns differ significantly: More evenly distributed departures compared to weekday concentration.
- Midday lull (11 AM - 3 PM) shows minimal activity at both airports.
- MCO shows more intense peak clustering than MIA—higher volume concentration during busy hours.

The dominant early morning peak reflects airline economics—aircraft that arrived the previous evening depart early to maximize daily rotations. MCO's stronger evening peak likely accommodates theme park visitors who spend full days at attractions before evening departures. The weekend spreading of departures suggests leisure traveler flexibility versus business traveler weekday time constraints. MIA's more distributed pattern may reflect international flight scheduling driven by connection timing rather than local preferences. The pronounced midday quiet period at both airports represents operational breathing room when airports can perform maintenance and reset operations between the morning and evening rushes. Understanding

these patterns allows airports to schedule staff, gates, and services to match actual demand rather than maintaining constant maximum capacity.

1.4. On-Time Performance Trend (≤ 15 min delay)

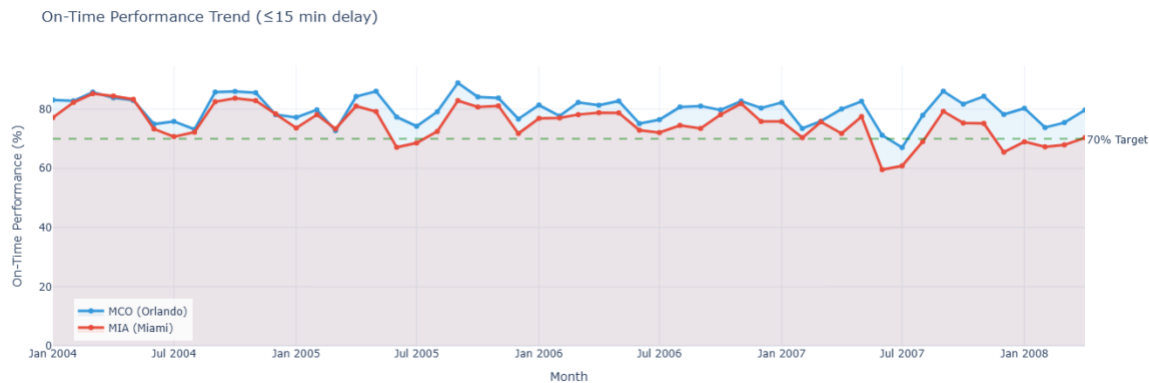


Figure 1.4: Comparison Of On-Time Performance Percentage Trend (≤ 15 min delay) for MCO & MIA (2004-2008).

On-time performance (OTP) measures the percentage of flights arriving within 15 minutes of their scheduled time, a key industry metric for operational reliability. This time-series analysis tracks monthly OTP rates to identify performance trends, seasonal variations, and crisis periods that affect passenger experience and airline competitiveness.

Key Findings:

- MCO maintains consistently better performance with ~79% average OTP compared to MIA's ~74%
- Both airports show declining trend from 2004-2007 reflecting system-wide aviation congestion
- Late 2007 shows dramatic performance collapse with MIA dropping to ~60% OTP
- Seasonal dips occur during summer months correlating with peak traffic and weather events
- Neither airport consistently meets the 80% industry target during the study period

The persistent 5-percentage-point gap between MCO and MIA reveals fundamental operational differences. Orlando's point-to-point leisure traffic model proves easier to manage than Miami's complex hub operations with international connections. When one international arrival delays at MIA, it cascades into multiple missed connections, compounding delay impacts. The 2007 crisis period coincides with record aviation system congestion nationwide plus severe summer weather—showing how external factors affect all airports but hit connection-dependent hubs harder. MCO's faster recovery in 2008 demonstrates the resilience advantage of simpler

operational models. For passengers, this data suggests Orlando offers more reliable arrival times, while Miami's hub complexity introduces inherent schedule risk despite the airport's strategic importance for international connectivity.

1.5. Day of Week Departure Patterns

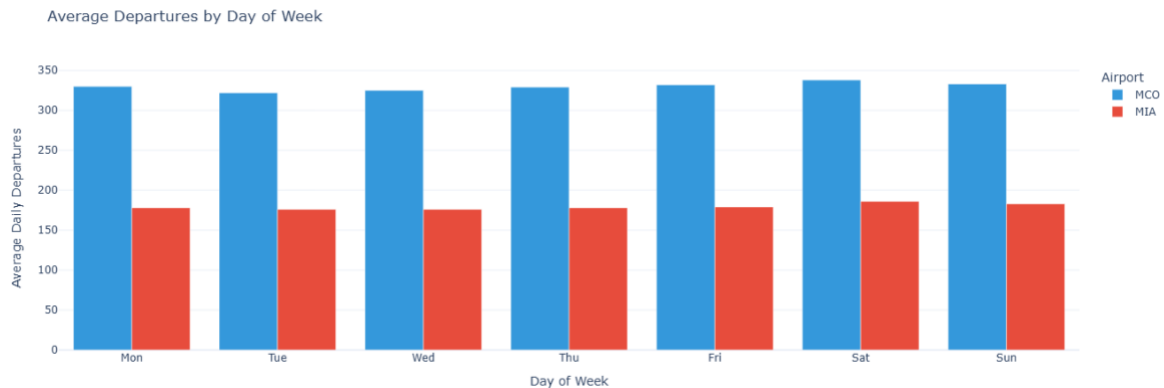


Figure 1.5: Comparison Of Average Departures by Day of the Week for MCO & MIA (2004-2008).

Weekly traffic distribution reveals the balance between business and leisure travel at each airport. Business-heavy airports show weekday peaks with weekend dips, while leisure-focused airports may show different patterns. Understanding these rhythms helps optimize staffing, pricing, and capacity allocation.

Key Findings:

- Friday is the busiest day at both airports with peak departure volumes
- Weekend traffic (Saturday-Sunday) is notably lower than weekday levels
- Monday through Thursday show relatively consistent volumes suggesting steady baseline demand
- Sunday shows recovery compared to Saturday, likely return travel from weekend trips
- The weekday/weekend gap is more pronounced at MCO indicating stronger business travel component

The Friday peak followed by weekend decline reveals a primarily business-oriented travel pattern overlaid with leisure demand. Business travelers departing Friday to return home create the weekly peak, while leisure travelers spread their departures across multiple days. The consistent Monday-Thursday baseline represents regular business travel and the steady flow of tourists arriving throughout the week. Sunday's recovery above Saturday levels captures the "return home" phenomenon as weekend leisure travelers head back. Interestingly, MCO shows this pattern despite being known for theme park tourism—suggesting that even Orlando attracts significant business travel for conventions, conferences, and corporate events. The pattern

indicates both airports should staff heavily for Friday operations while potentially reducing weekend capacity, though leisure demand prevents dramatic weekend cuts.

1.6. Average Arrival Delay by Month

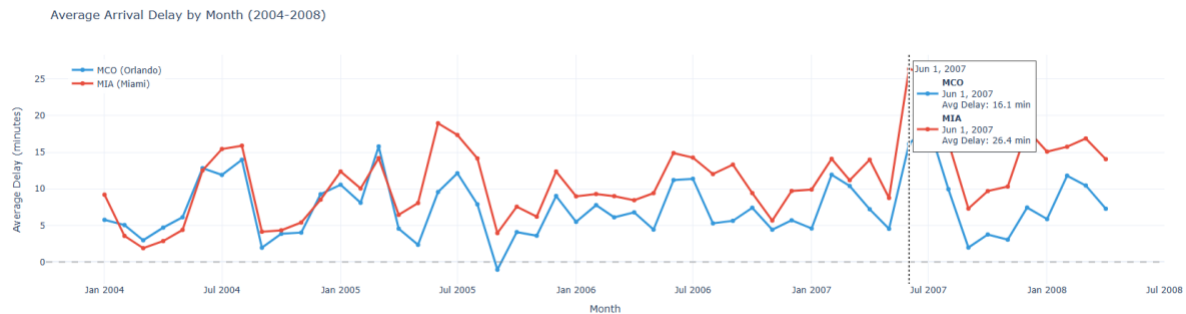


Figure 1.6: Comparison Of Average Arrival Delay by Month for MCO & MIA (2004-2008).

While on-time performance measures binary success (on-time or late), average delay quantifies the actual impact on passengers. This metric tracks mean arrival delays monthly to identify seasonal patterns, long-term trends, and periods of operational stress that affect passenger experience and airline costs.

Key Findings:

- MIA consistently shows higher average delays than MCO (11 minutes vs 8 minutes average)
- Summer 2007 shows dramatic delay spike with both airports exceeding 20-minute average delays
- Winter months generally perform better with delays near or below zero (early arrivals)
- 2008 shows improvement at both airports though data is limited to four months

The 3-minute average delay difference between airports may seem small but represents millions of passenger-hours annually given traffic volumes. MIA's higher delays stem from operational complexity—international arrivals face customs processing, longer taxi times, and greater exposure to connecting flight cascades. The summer 2007 crisis stands out dramatically, likely driven by that season's severe weather combined with peak traffic loads. Winter's better performance reflects both lower traffic volumes and more favorable weather conditions. The multi-year degradation from 2004-2007 mirrors national aviation trends as traffic grew faster than infrastructure capacity. Notably, negative delays (early arrivals) occur frequently in winter, suggesting airlines pad schedules during challenging seasons or benefit from favorable winds. The 2008 improvement, though based on partial data, may reflect the beginning of traffic reductions as the financial crisis reduced demand and eased system congestion.

2. Deep Dive in Delay Dynamics -- Two Airports, Two Systems

Between 2004 and 2008, Florida's two largest commercial aviation hubs—**Orlando International (MCO)** and **Miami International (MIA)**—followed sharply divergent operational trajectories. As we can see below, the daily mean gap delay (in minutes – MCO-MIA), became increasingly negative, showing an deterioration of MIA performance in comparison with MCO.

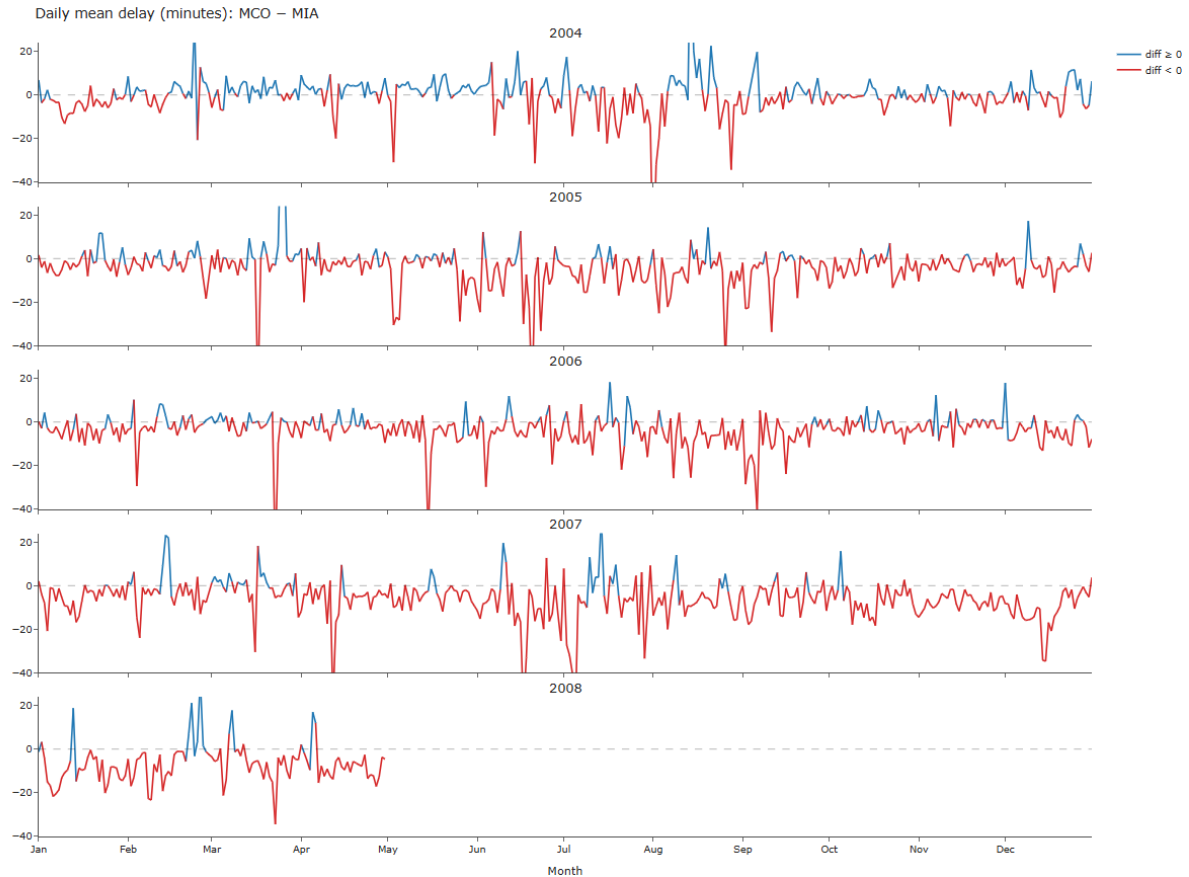


Figure 2.1: Gap (MCO-MIA) of Daily Average Delay (in minutes). Red corresponds to negative, meaning that MIA performs worse (2004-2008).

Despite serving the same state, overlapping carriers, and broadly similar macroeconomic conditions, the two airports exhibit fundamentally different delay dynamics.

The evidence shows that:

- MCO behaves like a stable, **capacity-managed system**, where delays are largely transient and scale modestly with traffic.
- MIA behaves like a fragile, **saturation-prone system**, where delays grow over time, respond nonlinearly to congestion, and propagate strongly across hours.

This divergence is not anecdotal. As we will see, it is visible consistently across:

- time trends
- congestion response curves
- hourly backlog persistence
- seasonal heatmaps
- airline-level contributions

The result is a widening reliability gap that cannot be explained by demand alone and instead reflects structural differences in recovery capacity and operational resilience.

2.1. Data and measurement

The primary outcome variable used in this section of the analysis is arrival delay (ArrDelay), measured in minutes. Arrival delay is selected as the main performance indicator because it more accurately reflects airport-level operational efficiency and congestion. Delays incurred at departure that are subsequently absorbed during flight (e.g., through speed adjustments or favorable en-route conditions) have limited impact on airport operations and passenger experience upon arrival, and are therefore not the focus of this analysis.

When analyzing delays, the dataset is restricted to completed flights, defined as flights that were neither cancelled nor diverted. Cancelled and diverted flights are excluded because arrival delay is not operationally comparable in cases where a flight does not arrive at its intended destination under normal conditions. Including such records would distort delay statistics and obscure meaningful comparisons between airports.

2.2. Long-Run Trend Divergence

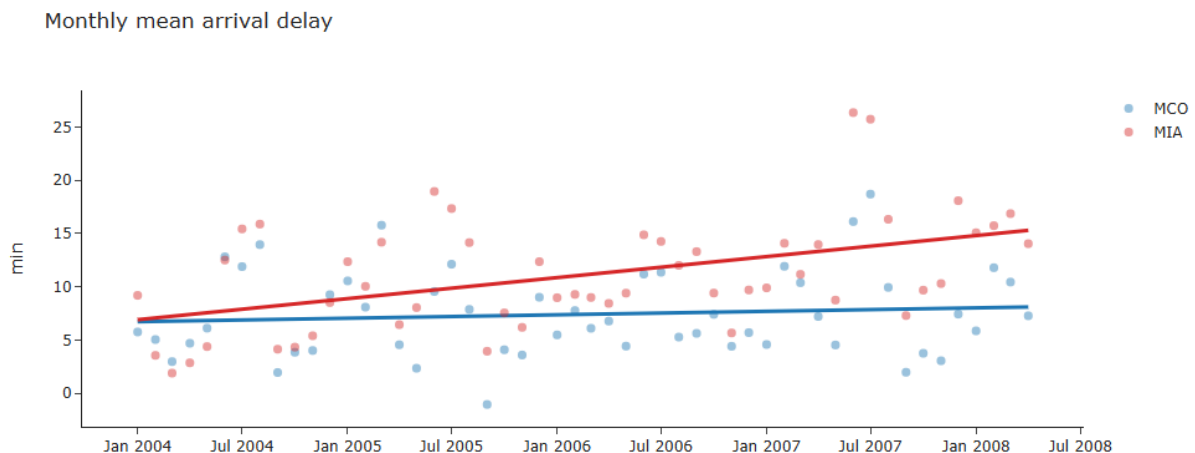


Figure 2.2: Trend Comparison of Monthly Average Arrival Delay Times (in minutes) for MCO & MIA (2004-2008).

We estimate monthly trends in mean arrival delay using HAC-robust inference.

- MCO:

- Trend $\approx +0.03$ min/month
 - Statistically indistinguishable from zero ($p > 0.5$)
 - Indicates **stationary performance**.
- MIA:
- Trend $\approx +0.16$ min/month
 - Highly significant ($p < 0.001$)
 - Implies a **compounding deterioration** in operational performance.

Interpretation:

The upward trend in mean arrival delays at MIA cannot be attributed solely to short-term seasonal fluctuations; instead, it suggests a persistent deterioration in operational performance over time. In contrast, MCO displays relatively stable, mean-reverting behavior, indicating greater resilience to transient congestion and disruptions.

2.3. Reliability Risk, Not Just Average Delay

While average delays differ, the **tail risk** differs far more.

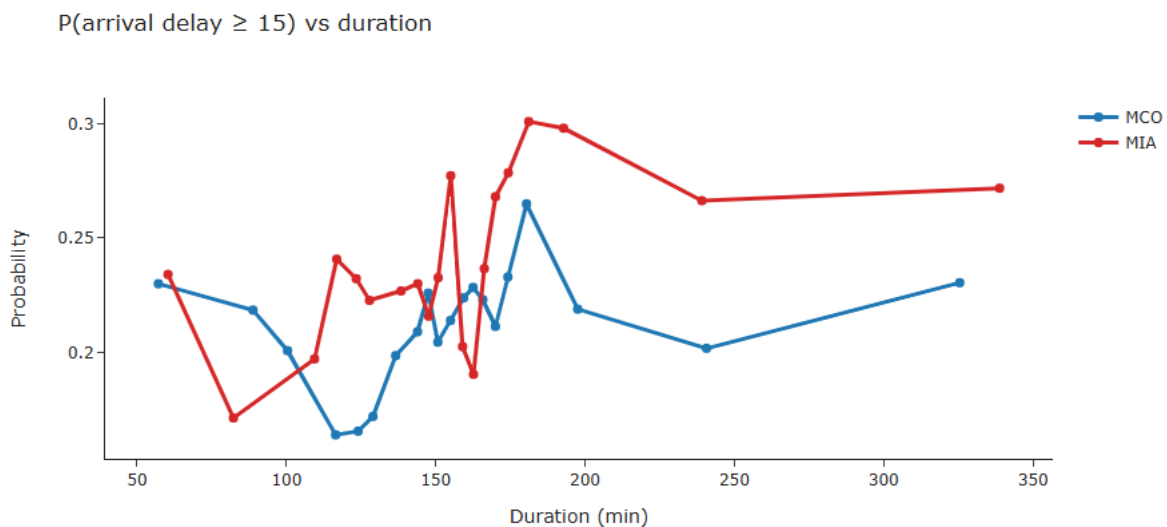


Figure 2.3: Probability of a flight being delayed (more than 15 minutes) in relation to the duration of the flight (in minutes) for MCO & MIA (2004-2008).

For flights exceeding ~ 150 minutes:

- $P(\text{ArrDelay} \geq 15)$ rises sharply at MIA,
- but remains approximately comparatively flat at MCO.

This asymmetry indicates that long-haul and inbound-dependent flights amplify existing congestion, especially at MIA, where buffers are insufficient to absorb shocks. This pattern is consistent with MIA's role as a major hub-and-spoke and international gateway, where delays on long inbound legs are more likely to cascade into arrival congestion. By contrast, MCO's flatter risk profile indicates greater robustness to delay propagation, potentially reflecting a higher share of point-to-point traffic and more effective recovery mechanisms.

2.4. Traffic Sensitivity and Congestion Regimes

Hourly traffic–delay analysis reveals two distinct regimes:

- MCO:

- Near-linear relationship between traffic and mean delay.
- Delays increase, but predictably and proportionally.

- MIA:

- Strong nonlinearity once traffic reaches $\sim 10 - 15$ flights/hour.
- Beyond this threshold, there are improvements.

Formally:

$$\text{Mean ArrDelay}_{\text{airport}} \approx a + \gamma \cdot \text{Traffic} + \delta \cdot l(\text{Peak})$$

Mean arrival delay vs traffic

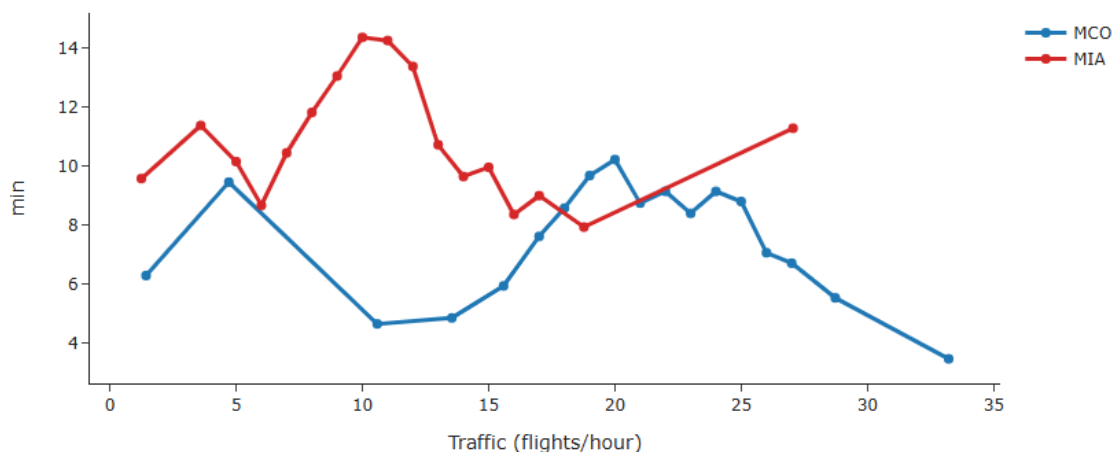


Figure 2.4: Average Arrival Delay (in minutes) of a flight in relation to the Traffic of the Airport (flights per hour) for MCO & MIA (2004-2008).

After conditioning on hour and day MIA incurs ≈ 4.6 additional minutes of delay relative to MCO at comparable traffic levels.

Interpretation:

The observed nonlinearity at MIA is characteristic of a **binding ground-capacity constraint**, such as limitations in gate availability, taxiway throughput, or ramp coordination, rather than airspace congestion in isolation. In such regimes, congestion effects compound rapidly once capacity thresholds are exceeded, leading to delay amplification and reduced recoverability. The peak delay time on MCO comes more gradually, and later than MIA, as we would expect taking into account the size difference of the two airports.

2.5. Temporal Risk Concentration (Heatmap Evidence)

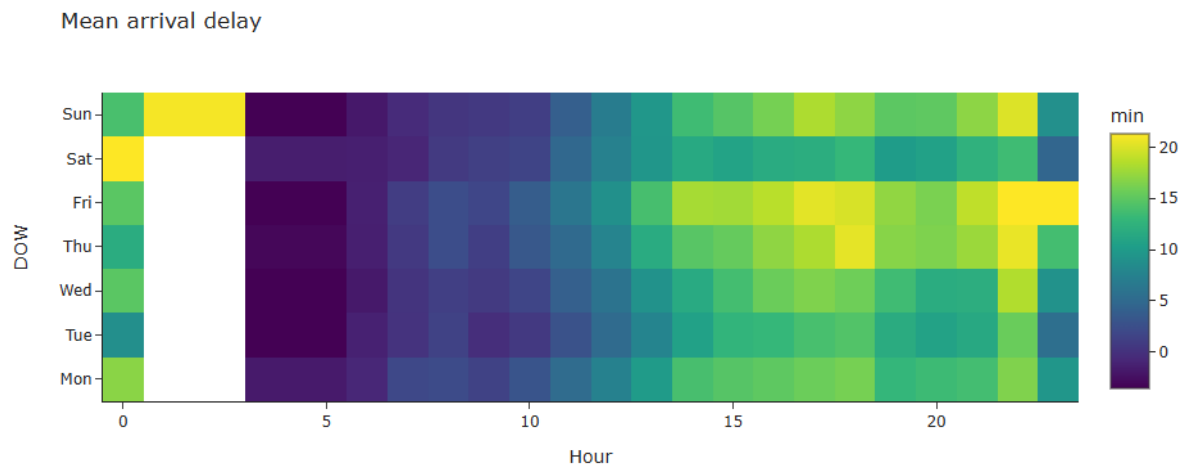


Figure 2.5: Heatmap (Hour x DOW) of Average Arrival Delay (in minutes) for MCO & MIA (2004-2008).

Heatmaps of arrival delay by hour, weekday, and month reveal systematic stress points:

- Midnight Backlog (Hour 0): Elevated delays reflect unresolved congestion carried over from the prior day.
- Hour-1 Void: A pronounced low-activity window (01:00) acts as a brief reset opportunity—but only if backlog is limited.
- Peak Fragility: Thursday–Friday afternoons (14:00–19:00) are the most failure-prone periods, especially at MIA.

These patterns indicate that delay accumulation, not only random shocks, drives poor performance.

2.6. The Summer Breakdown and Weather Amplification

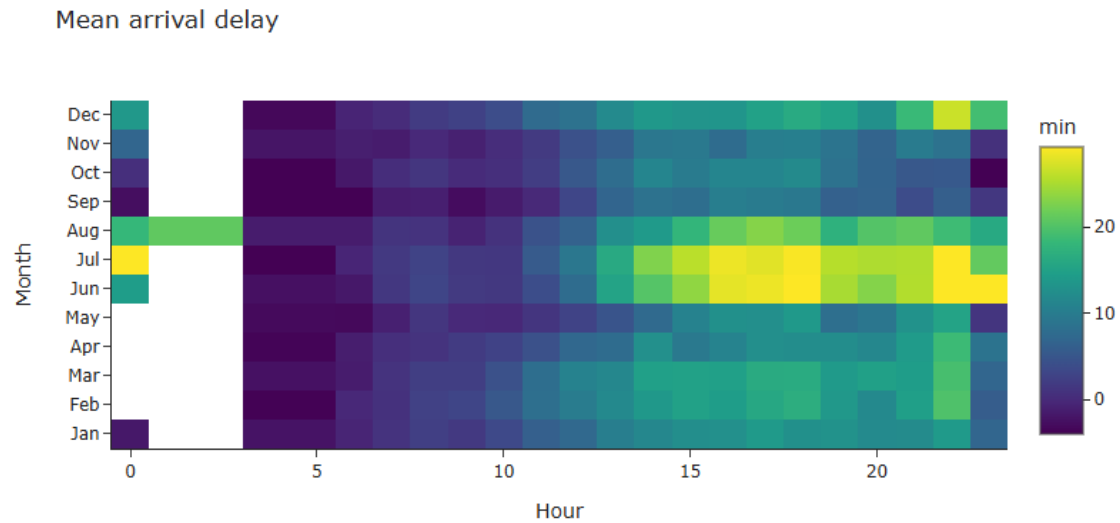


Figure 2.6: Heatmap (Hour x Month) of Average Arrival Delay (in minutes) for MCO & MIA (2004-2008).

Weather attribution analysis shows a sharp seasonal asymmetry:

- Weather-related delay share rises from ~2% to ~10% in June–August.
- This increase is not symmetric across airports.

At MIA:

- July evenings (18:00–22:00) frequently exceed 30 minutes mean arrival delay.
- Backlogs triggered by afternoon thunderstorms persist into late night.

At MCO:

- Weather shocks occur, but recovery is substantially faster.

Additional insight emerges when accounting for the seasonal composition of delay causes, as weather-related delays are predominantly concentrated in the summer months.

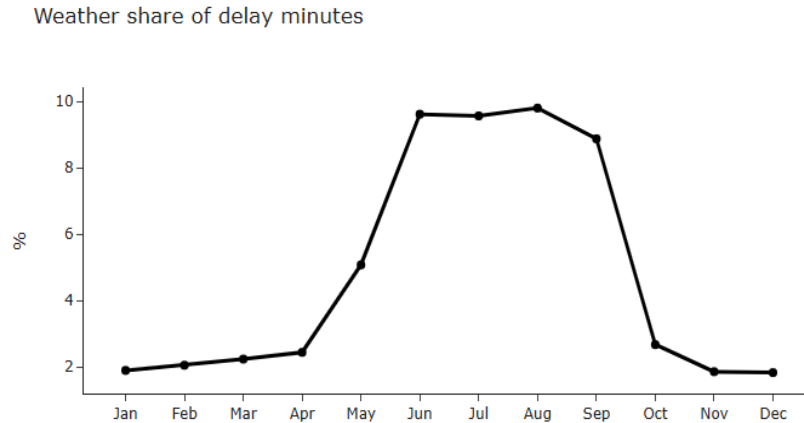


Figure 2.7: Total Weather Share (Contribution %) of delay minutes for MCO & MIA (2004-2008).

Interpretation:

Weather acts as a trigger, not the root cause. The real differentiator is recovery capacity—the ability to clear accumulated delays before the next operating cycle.

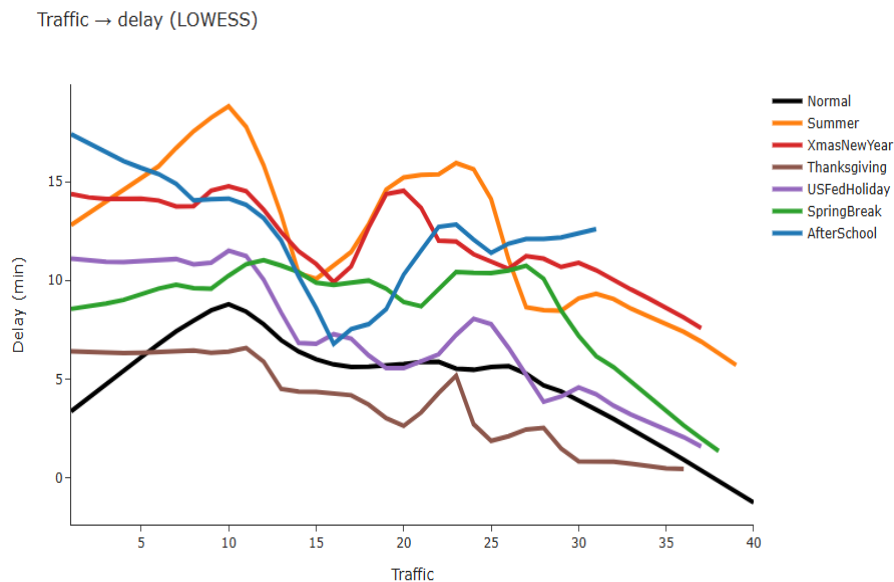


Figure 2.8: Traffic Delay (LOWESS Curves) by Occasion for MCO & MIA (2004-2008).

The LOWESS-smoothed relationship between traffic intensity and mean arrival delay reveals substantial seasonal heterogeneity in congestion dynamics. Rather than a single, stable traffic–delay curve, airport performance varies markedly across seasonal and holiday regimes, indicating that capacity constraints interact strongly with demand composition and external shocks.

During normal (non-peak) periods, delays increase gradually with traffic and remain relatively contained even at higher traffic levels. This pattern suggests that, under baseline conditions, the airport system operates with sufficient slack to absorb incremental demand without triggering severe congestion.

In contrast, summer months exhibit consistently higher delay levels across almost the entire traffic range. Notably, delays peak at moderate traffic levels (approximately 8–12 flights per hour) before declining at higher volumes, likely reflecting operational throttling mechanisms such as schedule smoothing, ground delay programs, or demand suppression during the most congested periods. The elevated summer delay curve aligns with the seasonal concentration of weather-related disruptions, which reduce effective capacity and increase recovery times.

Holiday periods display distinct and asymmetric effects. Christmas/New Year traffic is associated with elevated delays at low-to-moderate traffic levels, suggesting vulnerability to disruption even before nominal capacity limits are reached—potentially due to tighter schedules, crew constraints, and weather volatility. By contrast, Thanksgiving exhibits comparatively low delay levels across traffic intensities, consistent with highly concentrated travel peaks that airlines and airports may plan for more explicitly.

Intermediate regimes such as Spring Break, After-School, and US Federal Holidays occupy a middle ground, showing higher delay sensitivity than normal periods but lower systemic stress than summer. These regimes illustrate how demand surges tied to calendar-driven travel can partially erode operational buffers without fully overwhelming system capacity.

Interpretation

Overall, this figure demonstrates that traffic volume alone is an incomplete predictor of congestion. Instead, delay outcomes depend critically on when traffic occurs and under what operational conditions. Seasonal weather exposure, schedule rigidity, and passenger mix jointly shape the effective capacity of the airport system. In particular, the summer regime highlights how weather-induced capacity reductions can shift the traffic–delay relationship upward, amplifying congestion even at traffic levels that are manageable during other periods.

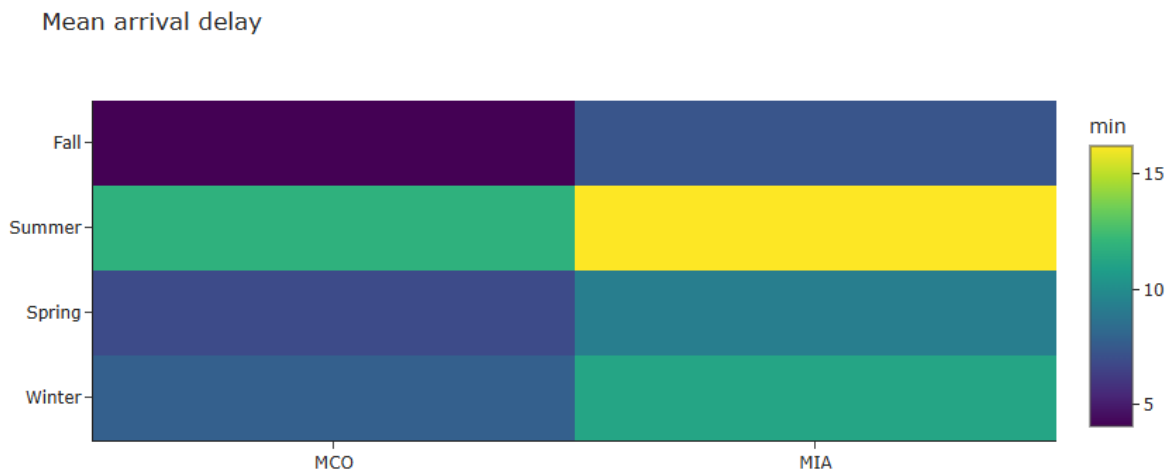


Figure 2.9: Heatmap (Airport x Season) of Average Arrival Delay (in minutes) (2004-2008).

The seasonal heatmap highlights pronounced differences in mean arrival delays across both airports and seasons, underscoring the role of seasonality in shaping operational performance.

For MCO, mean arrival delays remain relatively low and stable throughout the year, with the exception of summer, during which delays increase noticeably. The lowest delay levels are observed in fall, suggesting favorable operating conditions and sufficient capacity buffers during this period. Overall, MCO exhibits limited seasonal volatility, indicating a system that is comparatively resilient to seasonal demand and disruption shocks.

In contrast, MIA experiences consistently higher mean arrival delays across all seasons, with a particularly pronounced deterioration during the summer months, where delays peak substantially. Elevated delay levels persist into winter and spring, suggesting that congestion and disruption effects at MIA are not confined to a single season but reflect more persistent capacity constraints.

Interpretation

The stark summer amplification at both airports—especially at MIA—is consistent with the seasonal concentration of weather-related disruptions and elevated passenger demand. However, the magnitude of the summer effect at MIA suggests that adverse weather interacts with already tight operational margins, leading to disproportionate increases in delay. By contrast, MCO's more muted seasonal gradients indicate greater operational slack and a higher ability to absorb seasonal shocks.

2.7. The Pareto of System Pain — Who Contributes to Delay?

Delay minutes are highly concentrated among a small number of carriers.

Pareto: late-arrival minutes by airline

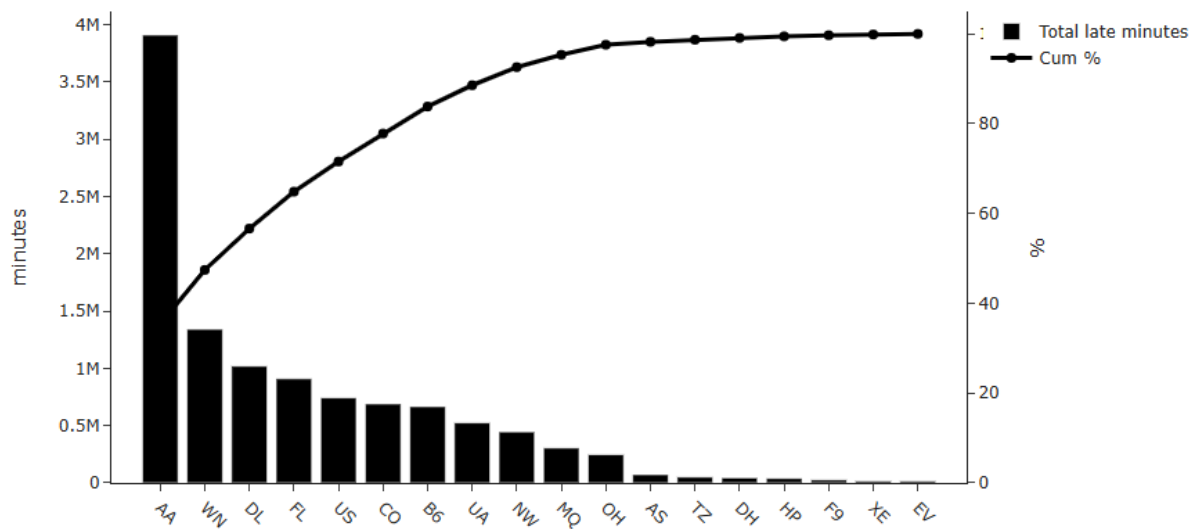


Figure 2.10: Pareto Plot (Cumulative Frequency & Declining in Histogram Frequency—min) of Top Airlines for MCO & MIA (2004-2008).

Carrier	Total Late Minutes	Share of Total	Strategic Role
AA	3.91M	~35%	Dominant hub carrier (MIA)
WN	1.34M	~12%	High-frequency operations
DL	1.02M	~9%	Network carrier
FL / B6 / US	—	—	Growth-oriented

The Pareto analysis of total late-arrival minutes by airline reveals a highly concentrated delay structure. American Airlines (AA) alone accounts for over one-third of total delay minutes, a magnitude that far exceeds that of any other carrier. The cumulative distribution indicates that a small subset of airlines—primarily AA, Southwest (WN), and Delta (DL)—is responsible for the majority of delay accumulation, consistent with a classic 80/20 concentration pattern.

Importantly, this concentration is not driven solely by differences in average delay severity. While AA also exhibits relatively high mean arrival delays compared to other carriers, the Pareto dominance is largely explained by its operational scale and hub dominance at MIA. As the primary hub carrier, AA operates a disproportionate share of flights, particularly during peak banks, amplifying the system-wide impact of any carrier-level inefficiencies.

The mean arrival delay distribution by airline further supports this interpretation. Several smaller or lower-frequency carriers exhibit comparable or even higher mean delays, yet their contribution to aggregate delay minutes remains limited due to lower traffic volumes. Conversely, AA's

combination of high frequency and above-average delay levels creates a compounding effect on airport-wide performance.

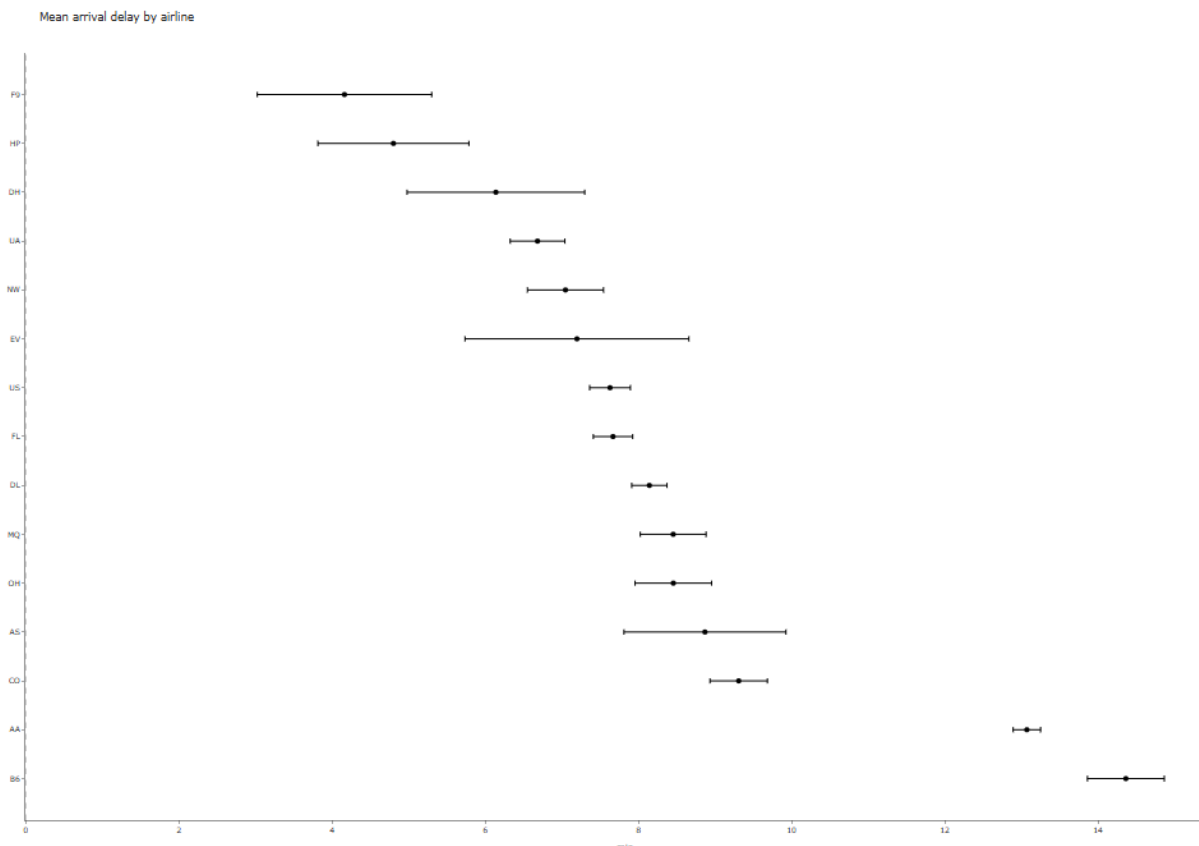


Figure 2.11: Mean Arrival Delay (in minutes) for the Top Airlines & 95% Confidence Intervals for MCO & MIA (2004-2008).

Taken together, these findings point to a carrier–airport coupling problem rather than an isolated airline performance issue. At a hub-dominated airport such as MIA, operational inefficiencies at the dominant carrier propagate rapidly through shared infrastructure, including gates, taxiways, and ramp resources. In turn, airport-level congestion further degrades the dominant carrier’s performance, creating a self-reinforcing feedback loop.

This dynamic implies that delay mitigation at MIA cannot be effectively addressed through airline-level interventions alone. Instead, it constitutes a hub governance challenge, requiring coordinated strategies between airport authorities, the dominant carrier, and air traffic management. Targeted measures—such as peak spreading, gate reallocation, or coordinated schedule smoothing—are likely to yield disproportionate benefits by addressing the structural source of delay concentration rather than its downstream symptoms.

2.8. Delay Persistence and Backlog Propagation (Core Structural Result)

Delays are not independent across time. Inbound aircraft, crews, and gates create temporal dependencies. To quantify this, we estimate:

$$\text{Mean ArrDelay}_t \approx a + \beta \cdot \text{ArrDelay}_{t-1} + \varepsilon_t$$

where β measures delay persistence.

Bellow we show LOWESS curves of current-hour vs previous-hour delay show:

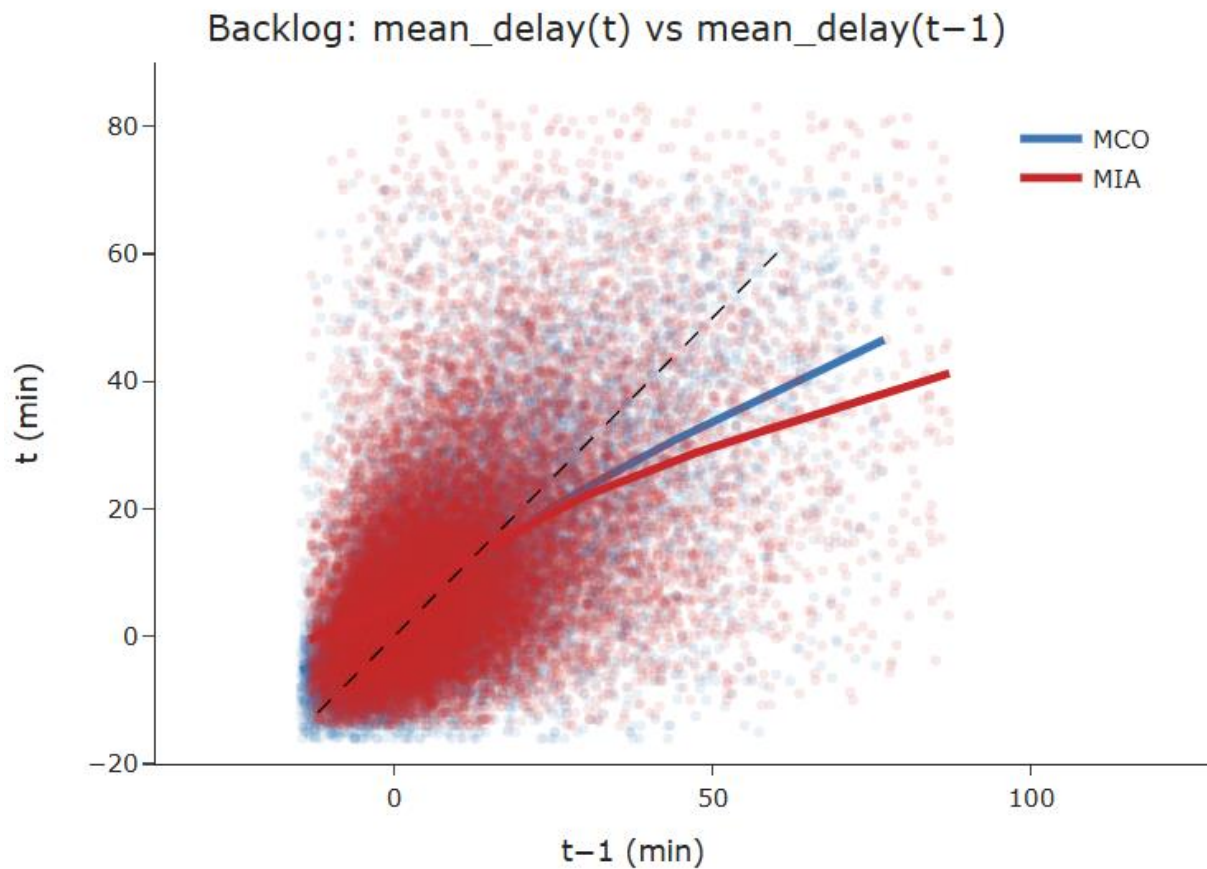


Figure 2.12: Backlog Plot: Mean Hourly Arrival Delay (in minutes) with respect to the Previous Hourly Arrival Delay for MCO & MIA (2004-2008).

Both airports exhibit dynamics that lie below the 45-degree ($y = x$) line, indicating that arrival delays are not fully persistent from one hour to the next and that some degree of recovery occurs over time. However, the extent of this recovery differs markedly between the two systems.

The estimated backlog curve for MCO is notably steeper and closer to the full-persistence benchmark, implying a higher degree of hour-to-hour delay carryover. Once delays materialize at MCO, they tend to propagate more strongly into subsequent periods, reflecting slower dissipation of operational backlogs.

By contrast, MIA exhibits a flatter response, suggesting weaker persistence and faster partial absorption of delays across consecutive hours. While MIA experiences higher average delays overall, these delays appear less tightly coupled across time, consistent with a system that, although frequently congested, clears backlog more rapidly once conditions stabilize.

Interpretation

These patterns indicate that the two airports face different congestion regimes. MCO's higher delay persistence is consistent with a system that operates close to capacity but lacks rapid recovery mechanisms once disruptions occur. In contrast, MIA's lower persistence—despite higher delay levels—suggests a system characterized by frequent shocks but more aggressive or externally imposed reset mechanisms (e.g., schedule banks, air traffic control interventions, or cancellation-based clearing).

Airport	Backlog rate (β)	95% CI
MCO	0.60	[0.58, 0.62]
MIA	0.46	[0.44, 0.48]

Interpretation:

- ~60% of MCO's delay persists into the next hour.
- ~46% at MIA.

The key distinction:

- MCO:

- High persistence conditional on shocks,
- but fewer and smaller shocks overall.

- MIA:

- Lower hour-to-hour persistence,
- but larger shocks, higher congestion sensitivity, and worse recovery after peak periods.

In short:

- MCO is a **stable system** with memory.

- MIA is an **unstable system with frequent resets**—but at a higher baseline.

2.9. Policy and Governance Interventions

2.9.1. Performance-Adjusted Landing Fees (PALF)

Proposal: Impose a 15–20% landing-fee surcharge on carriers in the bottom decile of hub-specific on-time performance.

- Encourages internalization of congestion externalities.
- Targets chronic contributors rather than all users.

2.9.2. Slot Recapture and Competitive Pressure

If a carrier:

- controls >50% of slots and
- contributes >30% of delay minutes,

then forced slot divestiture should be triggered. Redistributing even 5% of peak slots can materially reduce systemic risk.

2.9.3. Time-of-Use Congestion Pricing

Heatmaps clearly identify July evenings (18:00–21:00) as a failure zone. 3× landing fees during these windows would:

- flatten demand,
- incentivize schedule shifting,
- and reduce peak backlog formation.

2.9.4. Summer Resilience Buffers

Given weather-delay shares near 10% in summer:

- Mandate 20% longer turnarounds for long-haul arrivals.
- Invest in high-speed taxi exits and A-CDM systems.

2.10. Remarks about the delays

MCO and MIA are not just busier or quieter versions of the same system.

They are structurally different machines.

- MCO demonstrates controlled, predictable delay dynamics.

- MIA exhibits fragility, congestion amplification, and long-run deterioration.

The national map of mean arrival delays reveals a clear **spatial heterogeneity** in delay outcomes across the continental United States, indicating that airport performance is shaped not only by local conditions but also by broader **network connectivity and geographic exposure**.

Both **MCO and MIA** are located within a regional cluster of relatively elevated arrival delays in the southeastern United States. However, **MIA exhibits consistently higher delay levels than MCO**, placing it among the more delay-prone nodes in the national network. This suggests that MIA's performance challenges are not isolated but embedded within a broader set of high-delay origin and destination regions.

The concentric distance bands further illustrate the role of **long-haul connectivity**. Airports and counties located at intermediate to long distances from South Florida—particularly along major east–west and northeast corridors—tend to exhibit higher mean arrival delays. This pattern is consistent with earlier findings that **longer flight durations amplify delay risk**, especially for airports that rely heavily on inbound traffic from distant nodes.

Carrier dominance, represented by marker shapes, adds an additional layer of structure. Regions dominated by large network carriers—especially **American Airlines**—frequently coincide with higher delay intensities. This reinforces the interpretation that **hub-and-spoke network configurations**, while operationally efficient under normal conditions, can exacerbate delay propagation when disruptions occur.

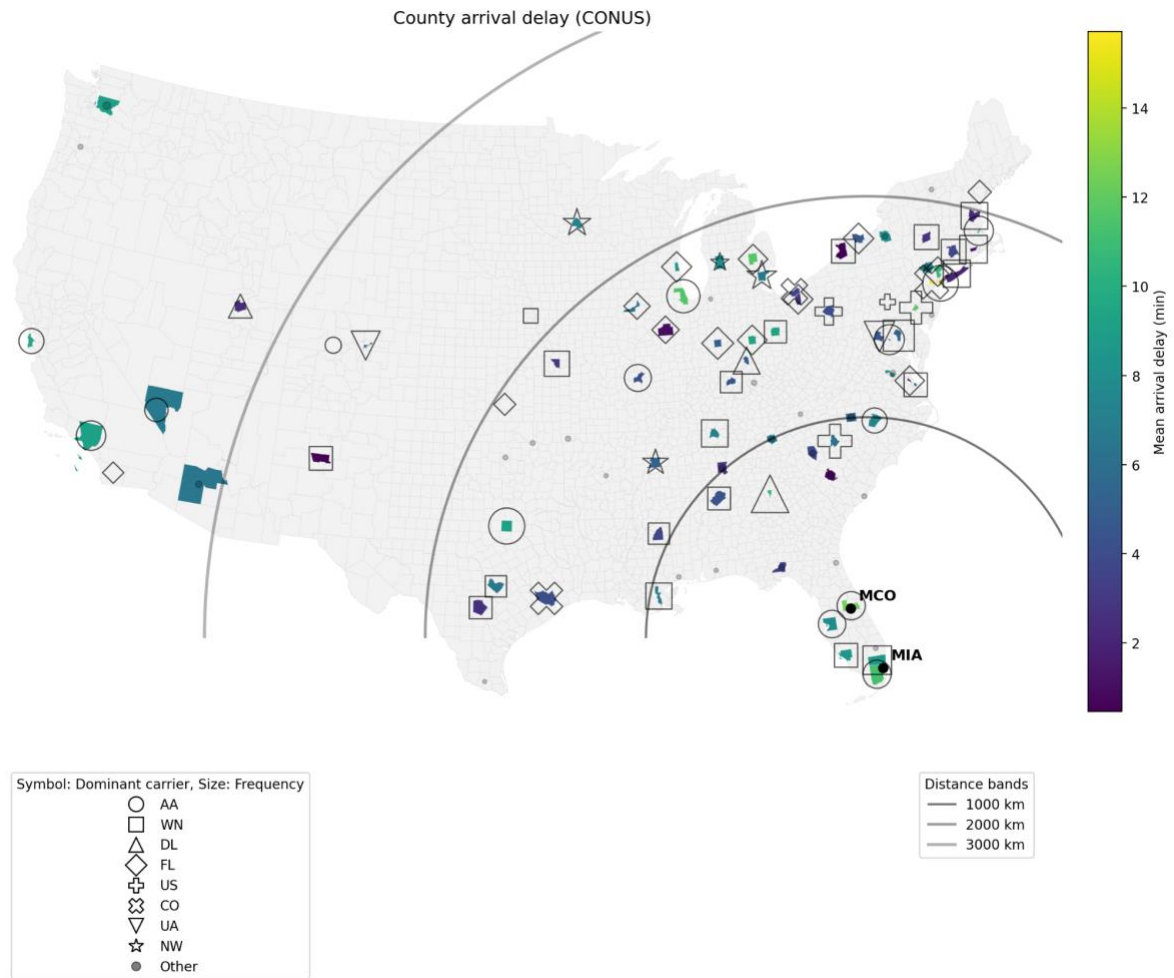


Figure 2.13: County Arrival Delay where color represents the average minutes of delay, Symbol the main airline & size the number of flights for MCO & MIA (2004-2008).

Interpretation

Taken together, the spatial distribution of delays highlights the importance of **network-level interdependence**. MIA's elevated delay performance appears to reflect not only local capacity constraints but also its strong coupling to distant, delay-prone regions within the national air transportation system. In contrast, MCO—while exposed to similar regional conditions—appears less embedded in high-delay long-haul corridors, contributing to its comparatively better performance.

The lesson is clear:

Airport performance is governed less by volume than by structure, incentives, and recovery capacity.

However:

Do delayed KPIs show the whole story?

Short answer: No. We need to take into account other metrics as the number of cancelled flights.

2.11. 9/11 and Airport Delay Performance: MCO vs MIA (2001)

This section evaluates whether the operational disruption associated with September 11, 2001 corresponds to a structural change in arrival delay behavior at Orlando International Airport (MCO) and Miami International Airport (MIA). The analysis combines micro-level and macro-level evidence to assess both distributional and time-series effects.

Two complementary strategies are employed:

- Flight-level pre/post analysis (micro evidence)
 - Comparison of arrival delay distributions (*ArrDelay*, in minutes) before and after 9/11
 - Restricted to completed flights only
 - Excludes the FAA ground stop period to avoid conflating grounded and resumed operations
 - Designed to detect changes in:
 - central tendency,
 - dispersion, and
 - tail behavior of delays
- Interrupted Time Series (ITS) / segmented regression (macro evidence)
 - Aggregation to weekly mean arrival delays
 - Estimation of a segmented model with a discrete break at the post-9/11 restart date
 - Allows identification of:
 - an immediate level shift, and
 - a potential post-event trend (slope) change

These approaches are intentionally complementary: flight-level tests provide high power for detecting distributional change, while ITS offers a transparent interpretation of structural breaks in the time series.

2.11.1. Flight-level results (completed flights, 2001 pre vs. post)

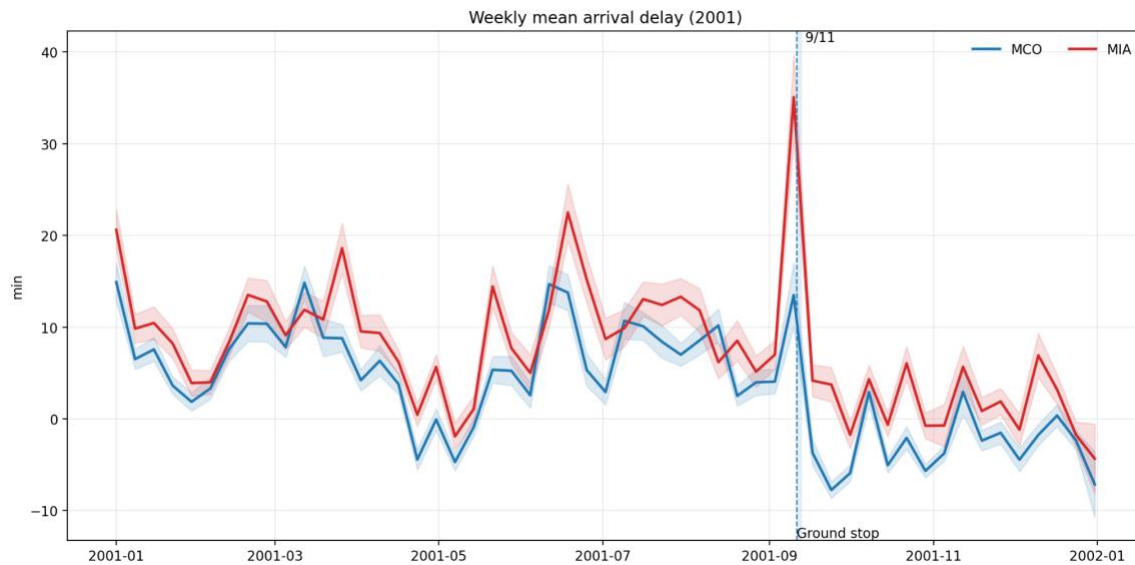


Figure 2.14: Weekly Mean Arrival Delay (in minutes) & 95% CI for MCO & MIA in year 2001 (Post & Pre 9/11).

Both airports experience substantial improvements in arrival delay performance after 9/11:

- MCO
 - Mean arrival delays decrease sharply
 - Strong compression of the upper tail:
 - 95th percentile falls from ~62 to ~27 minutes
 - Dispersion declines markedly, indicating more stable outcomes
- MIA
 - Mean arrival delays decrease by approximately 6.9 minutes
 - Upper-tail delays tighten:
 - p95 declines from ~75.9 to ~50 minutes
 - Overall variability also decreases

Multiple statistical tests confirm these changes for both airports:

- Mann–Whitney U: strong evidence of a shift in central tendency
- Kolmogorov–Smirnov (KS): significant distributional change (location, scale, or shape)

- Winsorized Welch t-test: robust mean reduction
- Levene (median-centered): significant reduction in variance
- Cliff's delta:
 - MCO: moderate effect size (≈ 0.23)
 - MIA: small-to-moderate effect size (≈ 0.16)

Both airports experience a similar post-9/11 drop in weekly mean arrival delays. The data do not support a statistically distinct “MIA response” relative to MCO in either level shift or slope change in the weekly series.

2.11.2. Interrupted time series results

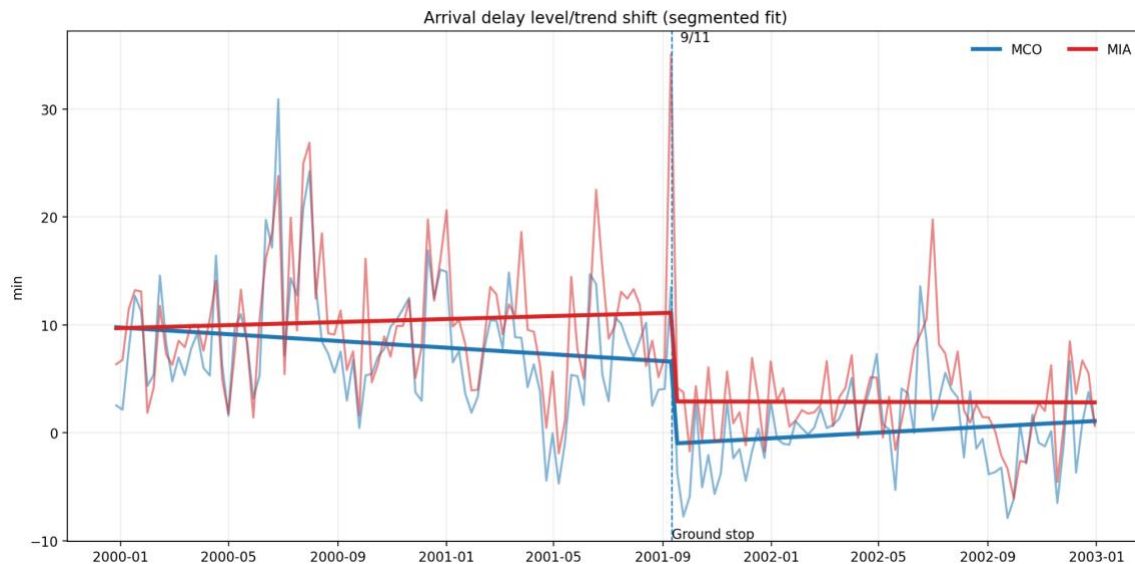


Figure 2.15: Weekly Mean Arrival Delay (in minutes) & Trend Shift for MCO & MIA in years 2000-2002 (Post & Pre 9/11).

The ITS analysis yields consistent conclusions:

- Both airports exhibit a large and immediate downward level shift in weekly mean arrival delays after operations resumed:
 - MCO: ~ 7.5 -minute reduction
 - MIA: ~ 8.2 -minute reduction
- Evidence for a post-event slope change is weak or absent
 - Indicates that the dominant effect is a level shift, not a sustained trend reversion

- A pooled weekly model finds:
 - no statistically significant difference between MCO and MIA in either level or slope changes

2.11.3. Delays versus cancellations

An important complementary finding emerges when cancellations are considered:

- **Cancellation rates increase sharply** in the post-9/11 period:
 - MCO: ~1.5% → ~3.1%
 - MIA: ~2.0% → ~3.9%
- This highlights that:
 - improved delay performance applies **conditional on flight operation**, and
 - cancellations represent a distinct operational adjustment margin

Bottom line

For both MCO and MIA the post-9/11 period shows a statistically and practically significant structural reduction in arrival delays among completed flights accompanied by a significant increase in cancellations. The ITS results indicate that the dominant change is an immediate level shift rather than a sustained change in trend, and pooled weekly results suggest the magnitude of the break is broadly similar across the two airports. So a reduction in delayed KPIs doesn't necessarily mean an overall improvement in performance.

3. MCO & MIA: The Bigger Picture

In this section we examine hurricane impacts, seasonal patterns, growth trajectories, and competitive market dynamics using U.S. Department of Transportation flight data (2004-2008) combined with NOAA hurricane records. The analysis reveals how airport function-MCO as a leisure destination versus MIA as an international gateway-shapes every aspect of operations from seasonality to market structure. Key findings include counterintuitive hurricane impact patterns, opposing seasonal cycles, and starkly different competitive landscapes.

3.1. Hurricane Timeline Impact Analysis (2004-2005)

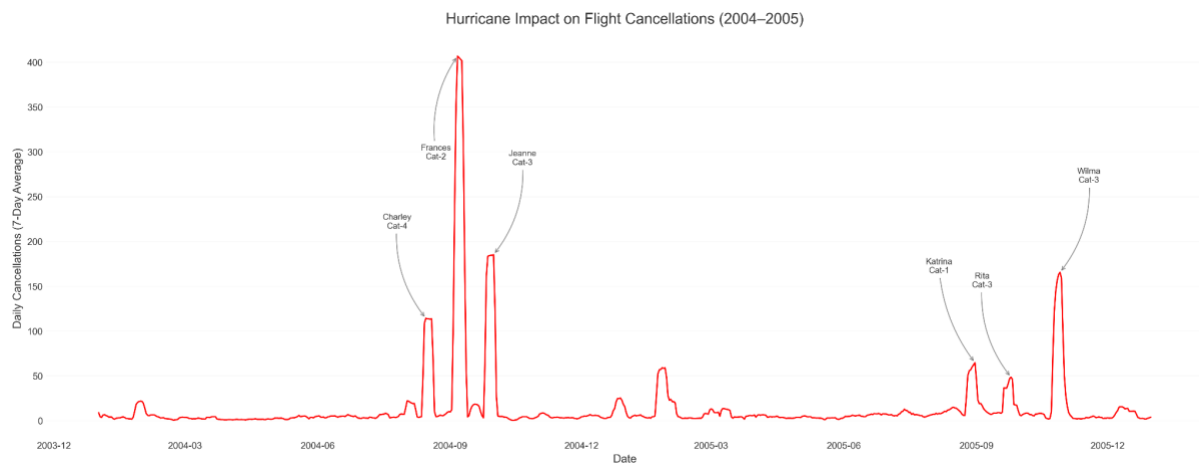


Figure 3.1: Timeline of major hurricanes affecting Florida during 2004-2005, showing daily flight cancellations. Hurricane Frances (Category 2) produced the highest single-day impact despite lower intensity than other storms.

Context: The 2004 Atlantic hurricane season saw four major hurricanes strike Florida within six weeks, creating unprecedented operational challenges for airports and airlines. The visualization tracks hurricane landfall dates against flight cancellation data to understand the relationship between storm characteristics and operational disruption.

Key Findings:

- Four major hurricanes (Charley, Frances, Ivan, Jeanne) struck within a six-week period in 2004 (August 13 - September 26)
- Hurricane Frances (Category 2, September 5) caused the worst single-day impact with 957 flight cancellations
- Total 2004 season resulted in 5,040 cancellations compared to 4,600 in 2005
- Hurricane category does not directly correlate with operational impact—Frances (Cat-2) exceeded Charley (Cat-4)
- Temporal clustering of storms compounds operational challenges as airports struggle to recover between events

Analysis: The counterintuitive finding that Category 2 Frances caused more disruption than Category 4 Charley reveals that storm size and forward speed matter as much as wind intensity. Frances was a significantly larger storm with a broader wind field, affecting a wider area for longer. This has critical implications for airport emergency planning—multi-factor assessment models incorporating storm size, path, and speed alongside intensity would provide more accurate operational risk assessments than category alone.

3.2. MCO vs MIA Hurricane Impact Comparison

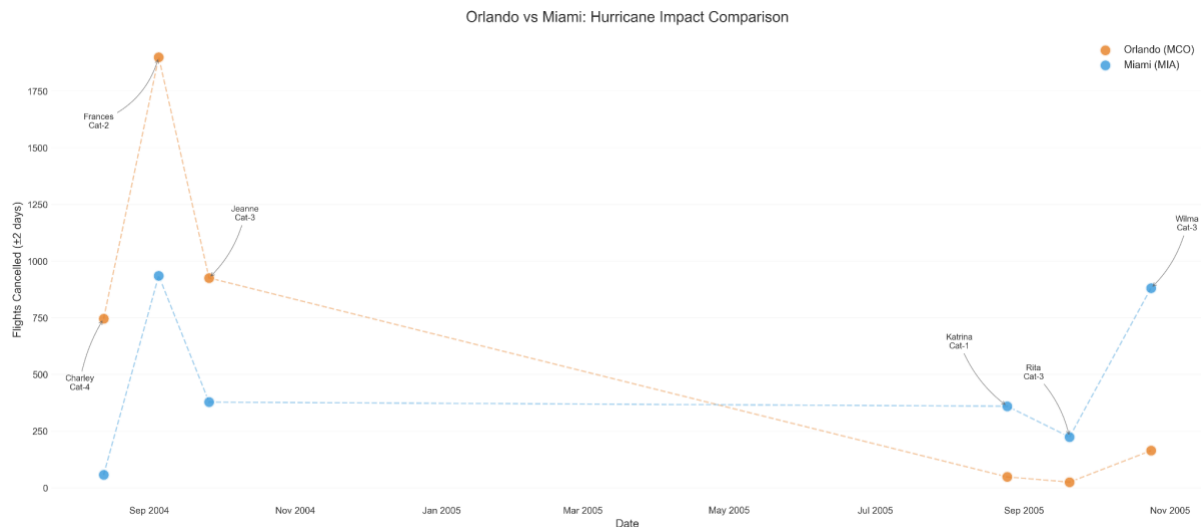


Figure 3.2: Comparison of hurricane impacts on Orlando (MCO) and Miami (MIA) airports showing distinct vulnerability patterns based on storm trajectory.

Context: Geographic analysis reveals how hurricane paths affect different airports asymmetrically. While MCO and MIA are both in Florida, their locations create different vulnerability profiles depending on specific storm trajectories.

Key Findings:

- MCO experienced severe impacts in 2004, particularly from Hurricane Frances (approximately 1,900 cancellations)
- MIA was disproportionately affected in 2005 by Hurricanes Katrina, Rita, and Wilma
- Hurricane trajectory and direct path impact operations more than storm intensity
- 2004 was "MCO's year" while 2005 hit MIA harder
- Airport-specific contingency planning must account for regional hurricane paths, not just general severity

Analysis: The differential impact patterns demonstrate that geographic position determines vulnerability more than overall storm strength. Frances tracked directly over Orlando in 2004, while the 2005 storms affected Miami more severely. This means each airport requires tailored contingency plans based on historical trajectory patterns rather than generic hurricane preparedness protocols. The one-size-fits-all approach to hurricane planning fails to account for these geographic realities.

3.3. Seasonal Flight Patterns - MCO vs MIA

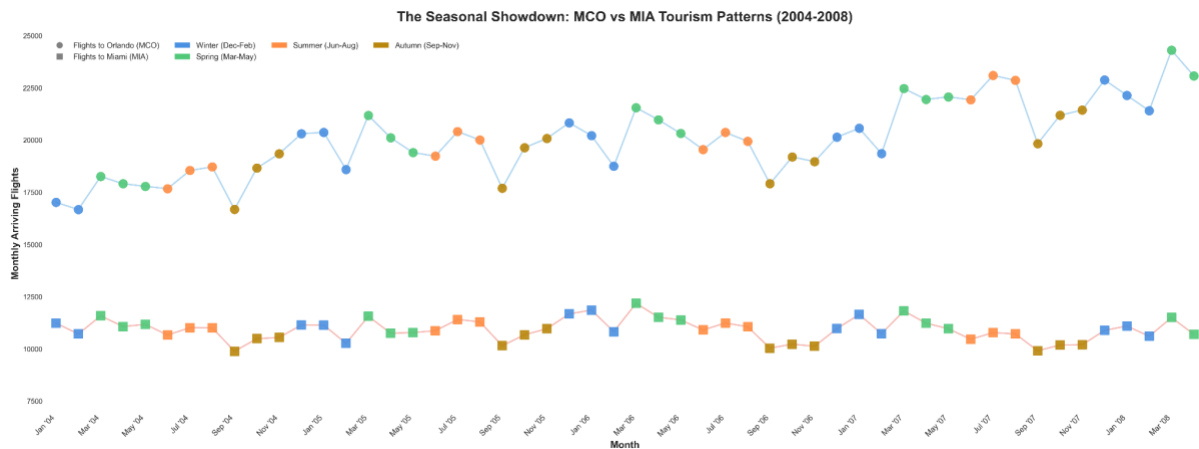


Figure 3.3: Granular monthly analysis quantifying seasonal variation and revealing September as the minimum traffic month for both airports.

Context: This detailed monthly breakdown quantifies the seasonal patterns and identifies common trends across both airports despite their opposite peak seasons.

Key Findings:

- MCO experiences approximately 40% increase in flight volume during summer months (June-August)
- MIA peak traffic occurs January-March, coinciding with peak winter tourism
- Both airports exhibit minimum traffic in September, correlating with Atlantic hurricane season peak
- September reduction suggests behavioral response to hurricane risk by passengers and airlines
- Pronounced seasonal variation creates operational challenges for workforce and capacity management

Analysis: The September minimum across both airports is particularly revealing—despite serving different markets with opposite seasonality, both experience the same low point during hurricane

season peak. This isn't coincidental; it reflects conscious risk avoidance by travelers and airlines. The 40% swing at MCO creates significant resource allocation challenges, requiring seasonal hiring and flexible capacity strategies that impact service quality and cost efficiency.

Top Destinations by Season

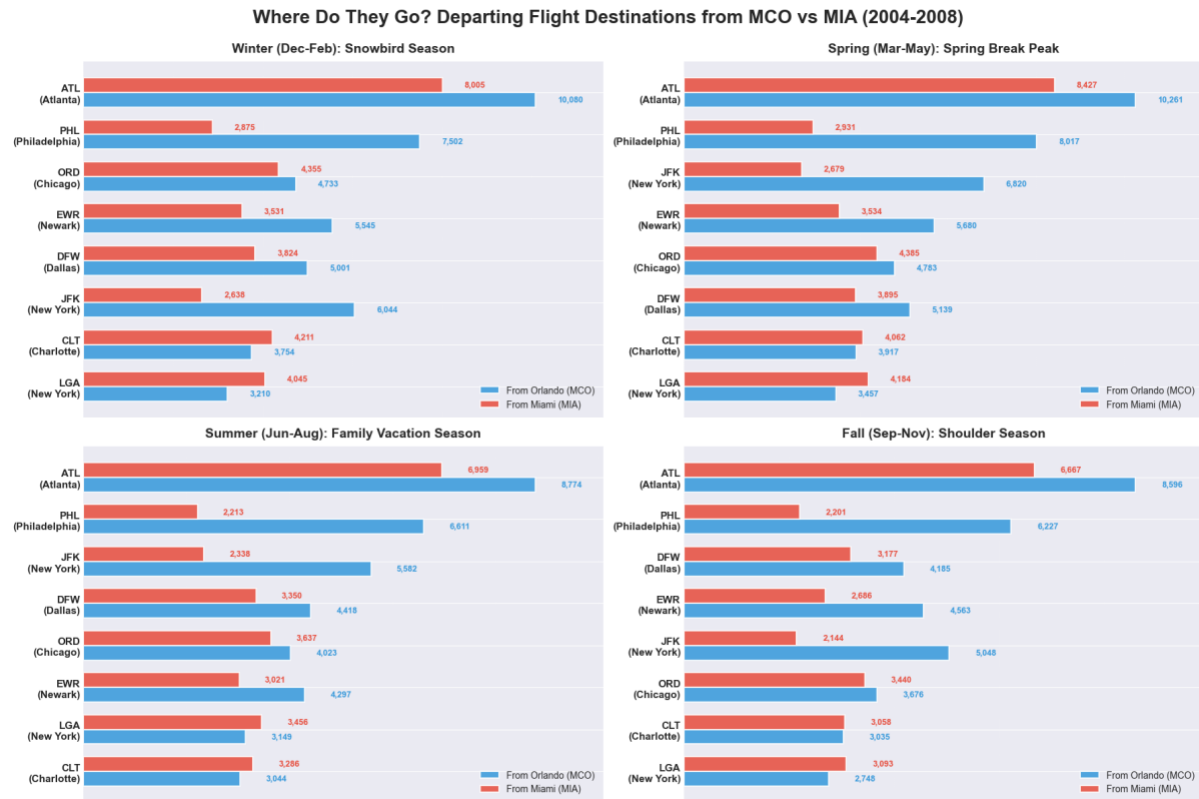


Figure 3.5: Seasonal destination analysis showing Atlanta's hub dominance and seasonal shifts between Caribbean and domestic routes.

Context: Route analysis identifies where passengers travel and how destination preferences shift seasonally, revealing network structure and market characteristics.

Key Findings:

- Atlanta (ATL) dominates as top destination from both MCO and MIA, reflecting Delta's hub power
- Seasonal destination shifts: Caribbean routes peak winter, domestic routes increase summer
- MCO demonstrates predominantly domestic route focus consistent with leisure market
- MIA exhibits international gateway characteristics with strong Latin America/Caribbean connectivity

- Hub-and-spoke structure creates asymmetric destination distributions with hubs capturing disproportionate shares

Analysis: Atlanta's dominance exemplifies hub power in the modern airline network—as Delta's primary hub, ATL captures traffic from both airports regardless of geographic logic. The seasonal shifts reveal market dynamics: winter drives Caribbean leisure travel from MIA, while summer increases domestic family travel. MCO's domestic focus versus MIA's international gateway role reflects their fundamental market positions. This network structure means hub disruptions have outsized impacts on the entire system.

The hurricane analysis reinforces the structural difference between MCO and MIA. While both airports face weather disruptions, MIA's hub-and-spoke model—with high traffic concentration and complex connection dependencies—makes it less resilient to external shocks. MCO's point-to-point model, while still affected, shows faster operational recovery post-hurricane, consistent with the delay persistence patterns observed in Section 2.8.

3.4. Distance vs Delay Performance

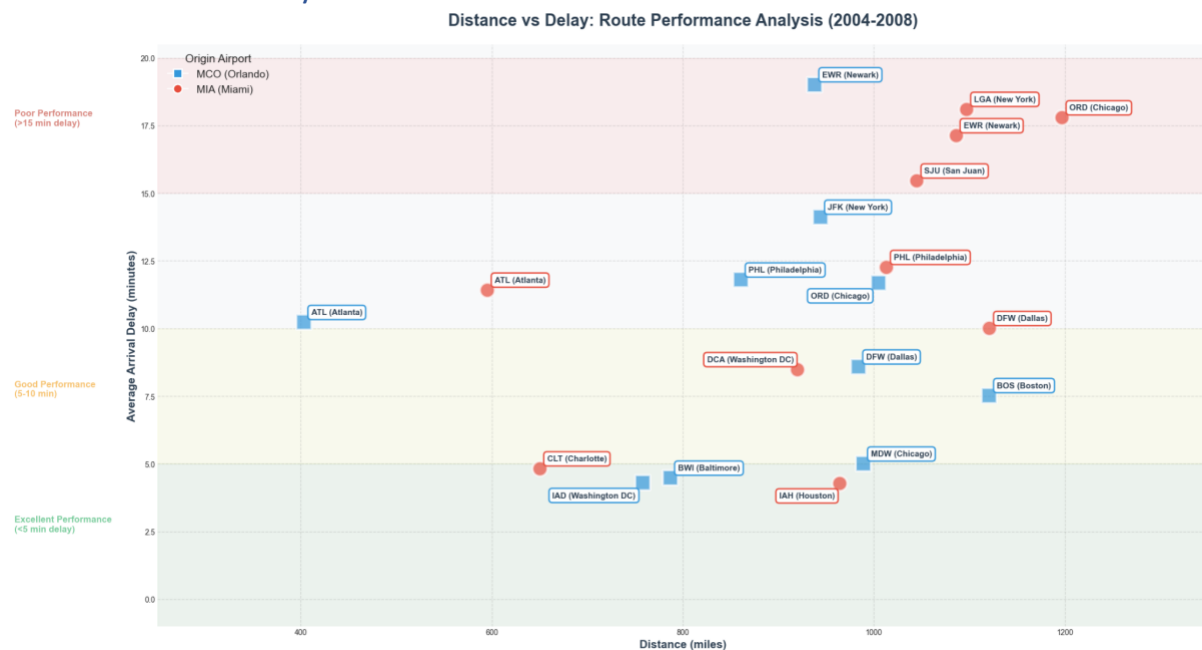


Figure 3.5: Counterintuitive relationship showing longer-distance flights achieve better on-time performance than short-haul operations.

Context: Conventional wisdom suggests longer flights face more delay risk, but operational reality tells a different story. This analysis examines how flight distance correlates with delay rates.

Key Findings:

- Longer-distance flights demonstrate superior on-time performance compared to short-haul
- Flights under 500 miles exhibit highest delay rates due to Northeast corridor congestion and hub saturation
- Flights exceeding 1,500 miles show improved punctuality with reduced exposure to cascading delays
- Short-haul operations face tight turnaround requirements and cumulative delay propagation across daily rotations
- Hub congestion management is critical factor in overall network reliability

Analysis: This counterintuitive finding reflects fundamental operational differences. Short-haul flights operate multiple daily rotations, so one delay cascades across subsequent flights. They also concentrate in congested Northeast corridors and major hubs where arrival/departure slots are tight. Long-haul flights typically fly point-to-point routes with lower congestion and have longer scheduled times that absorb minor disruptions. This suggests hub congestion management and quick-turn efficiency are more critical to system reliability than managing long-haul operations.

3.5. MCO vs MIA Growth Comparison (2004-2008)

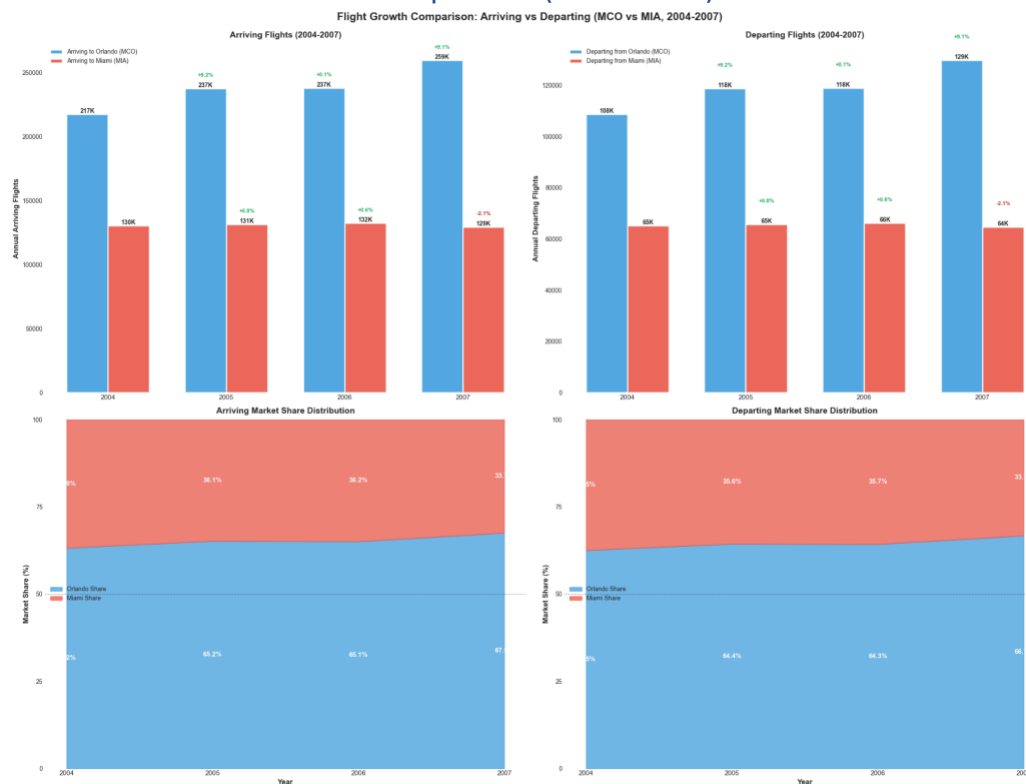


Figure 3.6: Longitudinal analysis showing MCO's explosive growth trajectory versus MIA's mature market stability, both impacted by 2008 recession.

Context: Growth trends reveal market dynamics and maturity. This five-year analysis tracks how each airport evolved during the mid-2000s boom and subsequent financial crisis.

Key Findings:

- MCO experienced approximately 25% growth from 2004 to 2007
- MIA exhibited minimal growth, consistent with mature hub market saturation
- Both airports experienced sharp traffic declines in 2008 coinciding with financial crisis
- MCO's growth reflects expanding theme park tourism and low-cost carrier market entry
- MIA's stability suggests mature market dynamics with limited organic growth capacity

Analysis: Orlando's explosive growth reflects the mid-2000s tourism boom and low-cost carrier expansion (Southwest, AirTran). As new attractions opened and budget airlines made Florida more accessible, MCO captured this expanding market. Miami, as a mature hub, had less room for organic growth—its market was already well-served with limited capacity for expansion within existing infrastructure. The 2008 crash hit both, but MCO's higher growth trajectory meant it had more momentum to lose. This illustrates how leisure markets show higher volatility than hub operations.

3.6. MCO Carrier Market Share Evolution

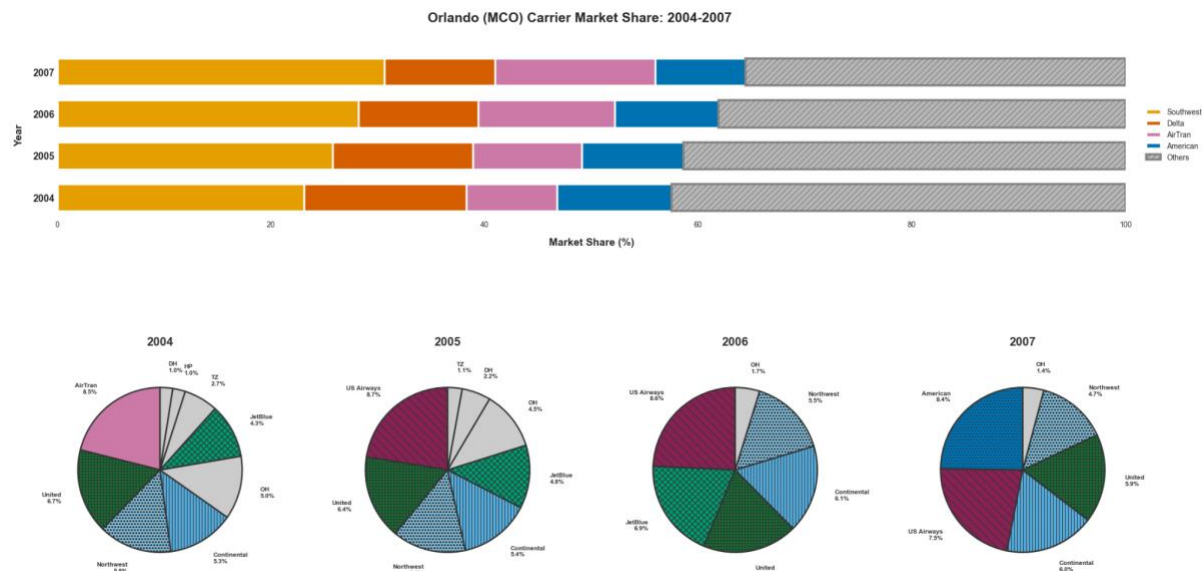


Figure 3.7: Market share analysis revealing Orlando as a competitive, tourism-driven market dominated by low-cost carriers, with detailed breakdown of the fragmented "Others" segment.

Context: Market structure analysis shows how carriers compete in the Orlando market and how this composition evolved over time.

Key Findings:

- Tourism-driven competitive market with significant low-cost carrier presence
- Top 4 carriers command majority share, with Southwest and AirTran (low-cost) dominating
- "Others" segment remains fragmented across multiple regional carriers
- Relatively low barriers to market entry indicated by viable competitive niches
- Visualization uses accessibility features (patterns, borders) for colorblind compatibility

Analysis: Orlando's market structure reflects its leisure destination status. Low-cost carriers thrive here because price-sensitive family travelers prioritize affordability. The fragmented "Others" segment shows the market remains competitive with room for regional carriers to capture niches. This contrasts sharply with hub-dominated markets where legacy carriers leverage network effects to maintain dominance. The competitive structure benefits consumers through lower fares but creates volatility as carriers enter and exit based on demand fluctuations.

3.7. MIA Carrier Market Share Evolution

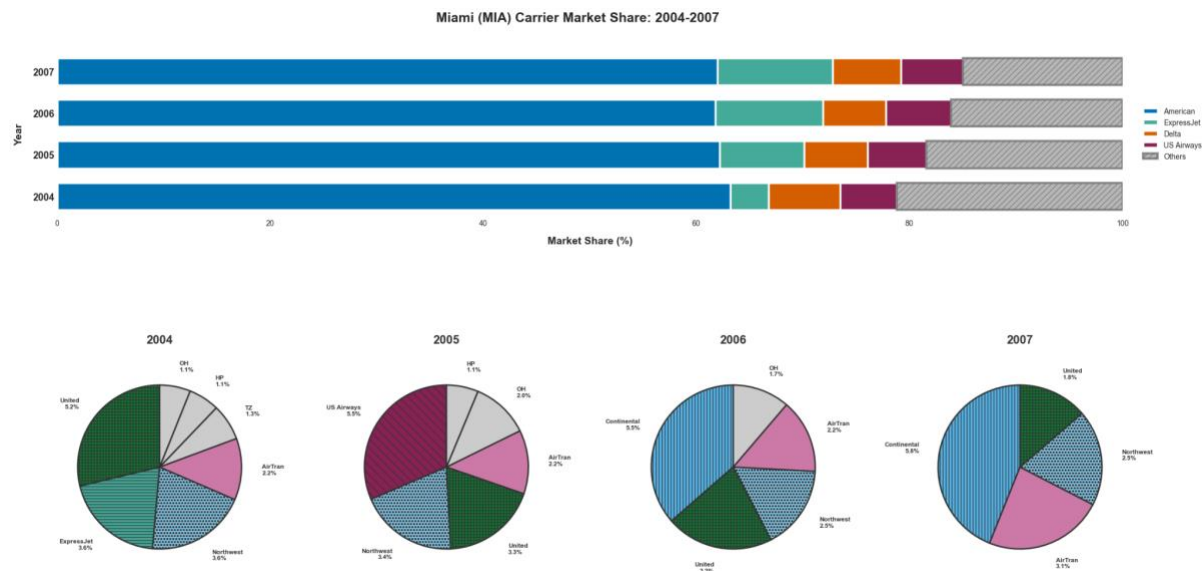


Figure 3.8: Miami market analysis showing hub-dominated structure with American Airlines' commanding position as Latin American gateway, contrasting with MCO's competitive dynamics.

Context: Miami's market structure reveals how hub dynamics create fundamentally different competitive environments compared to leisure destinations like Orlando.

Key Findings:

- Hub-dominated market with significantly higher concentration than MCO
- American Airlines demonstrates dominant position, leveraging MIA as primary Latin American hub
- Top 4 carriers control larger combined share than MCO equivalents
- Reduced "Others" segment suggests higher barriers to entry in hub-dominated markets
- Functional role (leisure vs hub) fundamentally shapes competitive dynamics

Analysis: Miami's concentrated market structure reflects American's strategic positioning of MIA as its Latin American gateway. Hub economics create natural barriers to entry—network effects, slot control, and infrastructure investments favor incumbent hub operators. The smaller "Others" segment compared to MCO shows how difficult it is for new entrants to compete against established hub operators. This MCO vs MIA comparison perfectly illustrates how airport function determines market structure: competitive leisure markets versus concentrated hub markets, each with different implications for pricing, service quality, and passenger choice.

4. EDA & Performance Tool

A dashboard that functions both as an analytical exploration environment as well as a performance review tool.

To better understand and familiarize with the flights data of the two airports, the following dashboard provided a consolidated analytical view. Once again, the period analyzed is from 01/01/2004 to 30/04/2008. The purpose is to support full operational diagnostics based on specific business questions which are answered by appropriate data visualizations. The image that follows describes this tool.

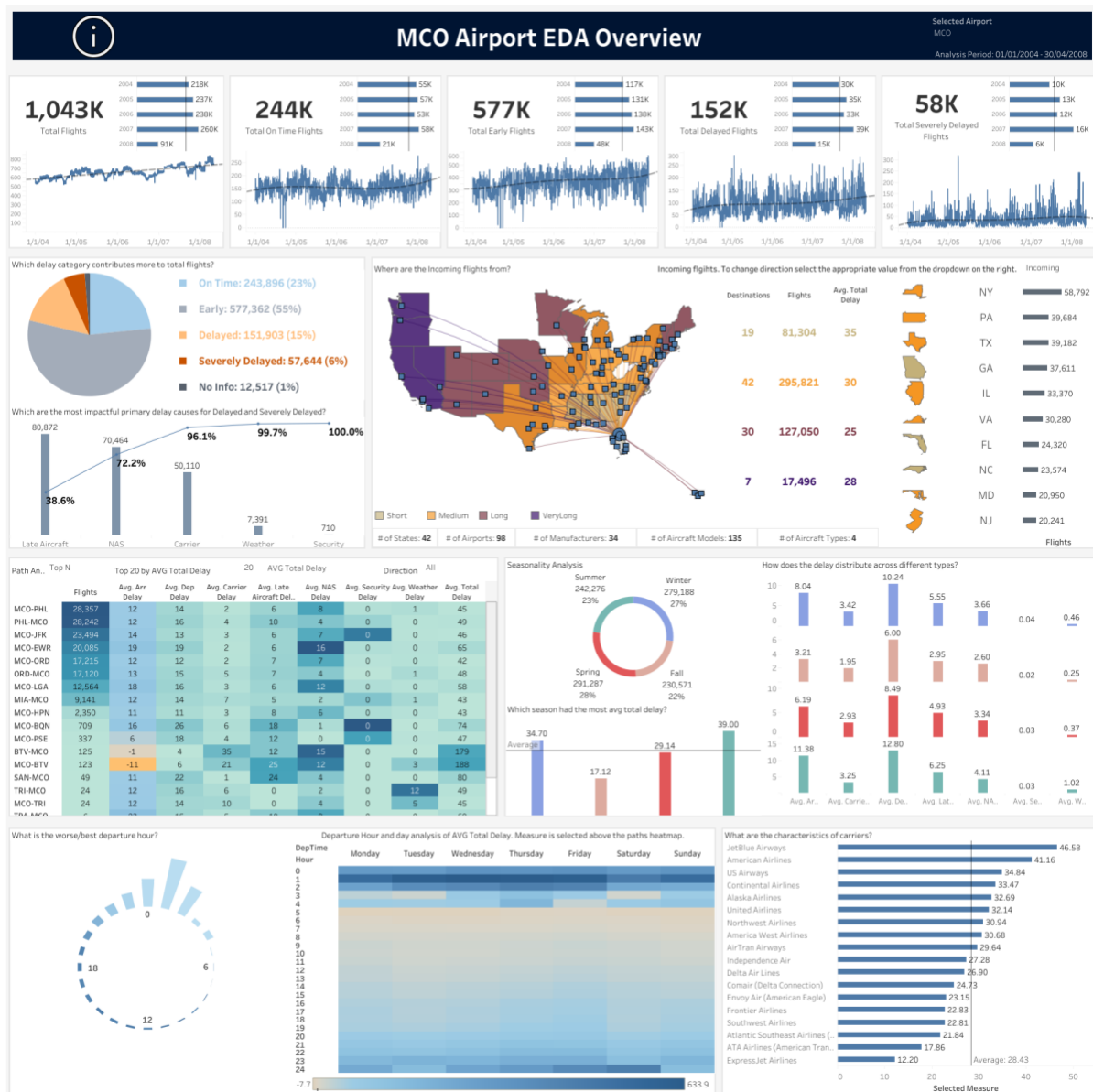


Image 4.1: EDA Overview Dashboard for selected Airport MCO. A dashboard that aims to provide value by creating an exploration environment as well as a performance review tool.

Insights are extracted when visualizations do not serve solely as pretty charts but as answers to specific questions posed by the business. The business questions addressed above regard traffic and reliability by answering questions such as what was the total flight volume, the distribution over time, punctuality and other. We also address delay structure and root causes for flights that were categorized as delayed or severely delayed. Relative questions are which are the primary reasons for delay and their contributions and which causes exist. Regarding Network and geography, specific visualizations help paint the picture of which routes the airport serves. Here, questions that are answered are which are the incoming / outgoing flights, how are flights spread into categories based on distance and others. Another core subject for flights is seasonality. To understand whether change in season impacts flights we provide a series of visualizations that drill down information from seasons down to departure hour. Lastly we also give context to Carriers and how they differ across different metrics.

The above dashboard empowers users to explore data, uncover underlying patterns and extract additional insights by being interactive. All visualizations are subject to change as users can select either MIA or MCO as airports for analysis. Apart from that, views can be customized from the user's actions. These actions include highlighting specific data, filtering to make the view more specific or choosing which metric should be analysed. Additionally, hovering over charts provides explanations about the visualizations and other information such as displaying values with more precision. In the following images a series of 4 interactive charts is presented. These charts were selected as they offer the most ways for the user to interact.

4.1. Dynamic Map Showing Incoming / Outgoing Flights:

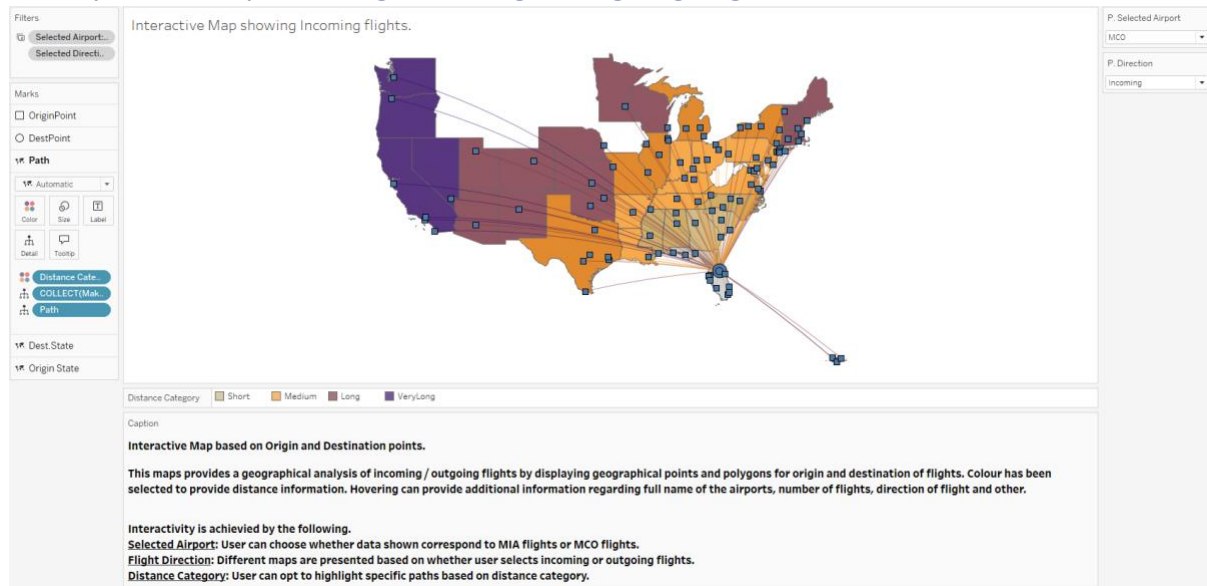


Image 4.2: Interactive Chart 1: Dynamic Map displaying geographical information regarding network.

This map provides a geographical analysis of incoming / outgoing flights by displaying geographical points and polygons for origin and destination of flights. Colour has been selected

to provide distance information. Hovering can provide additional information regarding the full name of the airports, number of flights, direction of flight and other.

Interactivity is achieved by the following.

- Selected Airport: User can choose whether data shown correspond to MIA flights or MCO flights.
- Flight Direction: Different maps are presented based on whether the user selects incoming or outgoing flights.
- Distance Category: User can opt to highlight specific paths based on distance category

4.2. Dynamic Day & Hour Heatmap:

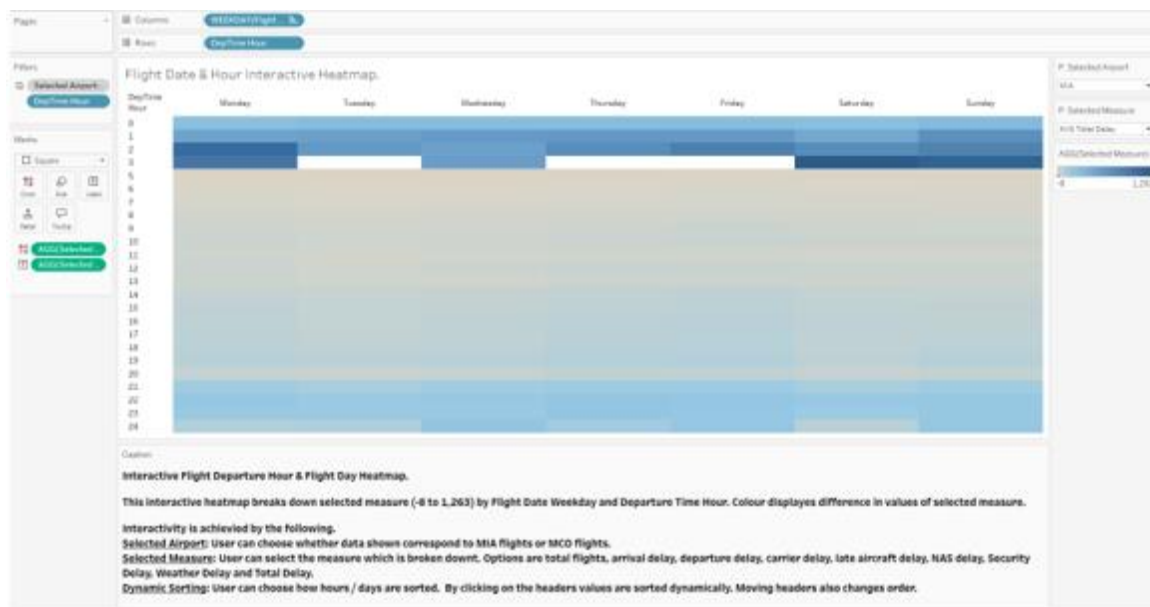


Image 4.3: Interactive Chart 2: Dynamic Heatmap aiming to provide information about the relationship of departure hour and flight date for specific metric.

This interactive heatmap breaks down selected measure by Flight Date Weekday and Departure Time Hour. Color displays difference in values of selected measure.

Interactivity is achieved by the following.

- Selected Airport: User can choose whether data shown correspond to MIA flights or MCO flights.
- Selected Measure: User can select the measure which is broken down. Options are total flights, arrival delay, departure delay, carrier delay, late aircraft delay, NAS delay, Security Delay, Weather Delay and Total Delay.
- Dynamic Sorting: User can choose how hours / days are sorted. By clicking on the headers values are sorted dynamically. Moving headers also changes order.

- Flight Direction: Different maps are presented based on whether the user selects incoming or outgoing flights.
- Distance Category: User can opt to highlight specific paths based on distance category.

4.3. Dynamic Carrier Horizontal Barplot:

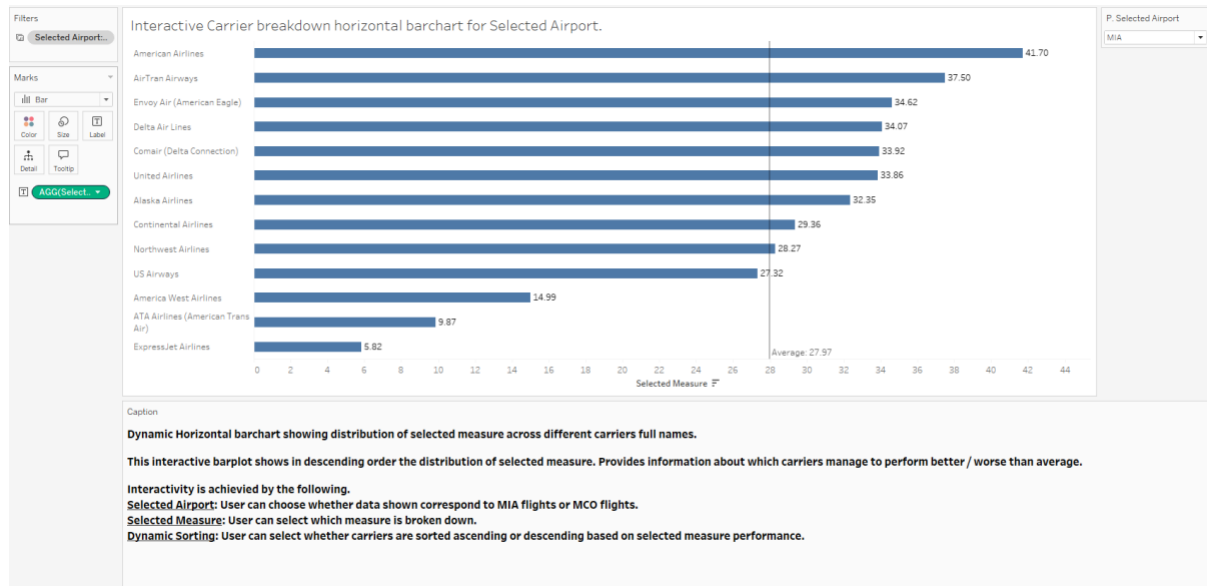


Image 4.3: Interactive Chart 3: Dynamic Horizontal barchart showing distribution of selected measure across different carriers full names.

This interactive barplot shows in descending order the distribution of selected measure. Provides information about which carriers manage to perform better / worse than average.

Interactivity is achieved by the following.

- Selected Airport: User can choose whether data shown correspond to MIA flights or MCO flights.
- Selected Measure: User can select which measure is broken down.
- Dynamic Sorting: User can select whether carriers are sorted ascending or descending based on selected measure performance.

4.4. Dynamic Paths Heatmap:

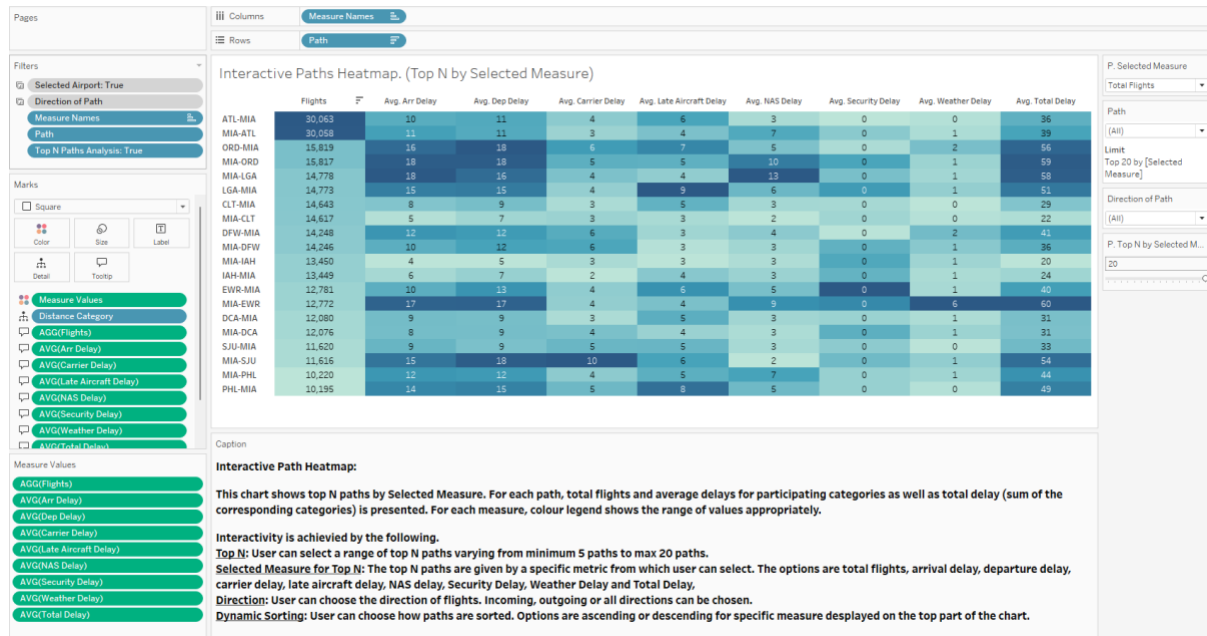


Image 4.3: Interactive Chart 4: Dynamic Horizontal barchart showing distribution of selected measure across different carriers full names.

The above visualization shows top N paths by Selected Measure. For each path, total flights and average delays for participating categories as well as total delay (sum of the corresponding categories) is presented. For each measure, colour legend shows the range of values respectively.

Interactivity is achieved by the following.

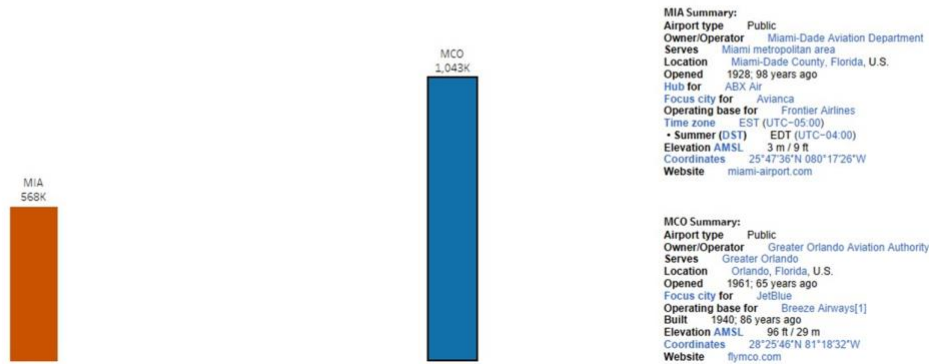
- **Selected Airport:** User can choose whether data shown correspond to MIA flights or MCO flights.
- **Top N:** User can select a range of top N paths varying from minimum 5 paths to max 20 paths.
- **Selected Measure for Top N:** The top N paths are given by a specific metric from which the user can select. The options are total flights, arrival delay, departure delay, carrier delay, late aircraft delay, NAS delay, Security Delay, Weather Delay and Total Delay,
- **Direction:** User can choose the direction of flights. Incoming, outgoing or all directions can be chosen.
- **Dynamic Sorting:** User can choose how paths are sorted. Options are ascending or descending for specific measure displayed on the top part of the chart.

4.5. Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA using Tableau Stories.

Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO handles twice the traffic	MCO exhibits lower cancellation rate despite traffic	MIA shows greater risk with more delayed flights	MIA outperforms MCO regarding On Time precision.	Different Risks are caused by different reasons	Distributional Statistics show greater risk for MIA.	MCO handles better busiest paths with lower delay.	Both Airports are affected by similar seasonal stress	Early departure hours show highest delay for both airports.
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For the same period of analysis (01/01/2004 - 30/04/2008), MCO handled nearly twice the total number of flights.

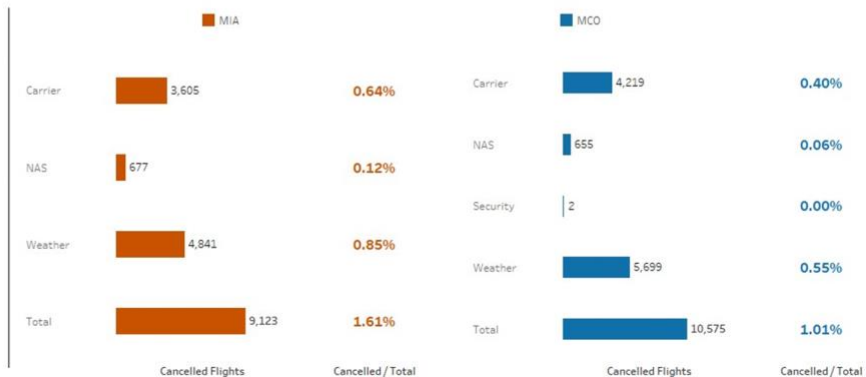


Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO handles twice the traffic	MCO exhibits lower cancellation rate despite traffic	MIA shows greater risk with more delayed flights	MIA outperforms MCO regarding On Time precision.	Different Risks are caused by different reasons	Distributional Statistics show greater risk for MIA.	MCO handles better busiest paths with lower delay.	Both Airports are affected by similar seasonal stress	Early departure hours show highest delay for both airports.
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Higher throughput does not inherently imply worse outcomes; MCO shows better handling regarding cancellation rates.

Despite handling significantly more flights overall, MCO exhibits a lower cancellation rate (1.01%) than MIA (1.61%), indicating lower relative operational reliability. Primary reason of cancellation for both airports is Weather.

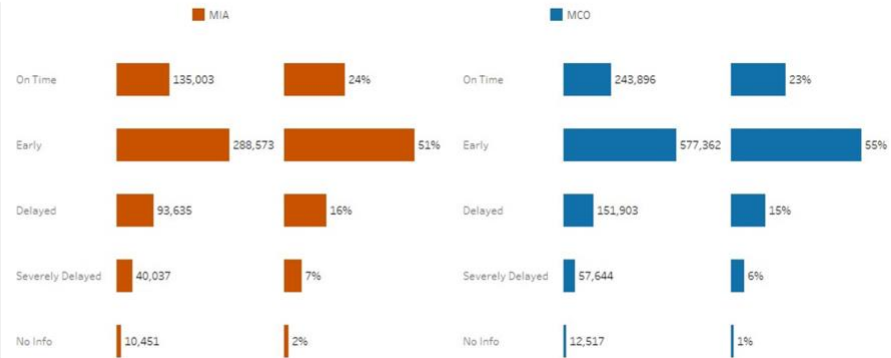


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Despite differences in traffic, punctuality is lacking for both airports with MIA showing higher risk.

Early flights are dominant for both airports showing lack of punctuality. On Time precision is nearly identical. Regarding delays, MIA shows 1p.p. difference both in Delayed and Severely Delayed flights indicating higher risk.

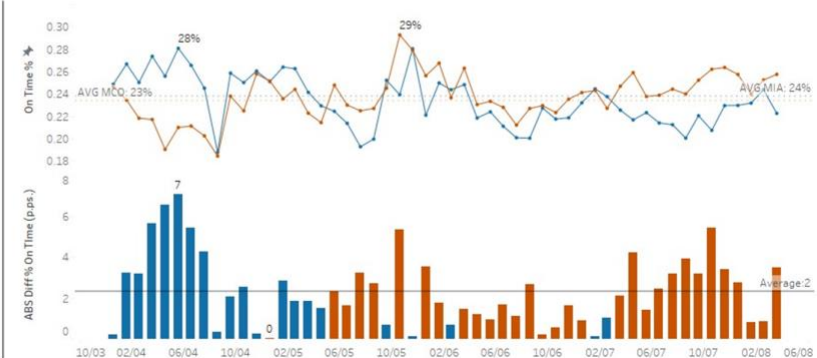


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MIA demonstrates higher reliability dominating in monthly On Time precision despite MCO better results in early period.

MIA outperforms MCO regarding On Time flights as indicated both by the higher average as well as the number of months with superior results. Common external factors affect the two airports as volatility shows similar fluctuations for both airports. Largest performance gaps are present in the early stages with later periods showing more stable differences and an average gap of approximately 2 percentage points. (Excludes flights regarding MCO-MIA or MIA-MCO)



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The two airports have different reasons for failure; MCO snowballs while MIA is prone to external dependencies.

Primary reason for delay (delayed & severely delayed) differs with NAS being the most significant reason for MIA while Late Aircraft is the most significant for MCO. Delays at MCO are more likely to snowball while MIA is more prone to external dependencies.



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MIA exposes passengers to higher risks as is indicated by comparing the distributional statistics.

MIA shows greater risk regarding delays with higher total average delay. Higher total average is accompanied by greater variability with IQR being more than 10 minutes greater. Both medians are around zero which shows that more extreme values are present in MIA.

Avg. Total Delay	Percentile (25) of Total Delay	Median Total Delay	Percentile (75) of Total Delay	IQR Total Delay	Variance of Total Delay	Std. dev. of Total Delay
38	-11	0	37	48	13K	115
30	-12	-1	25	37	10K	100

Total delay is giving by summing the minutes of delay across participating categories.

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Top 5 Busiest paths analysis backs MIA higher risk; MCO shows lower delays despite higher traffic.

MIA's highest-traffic routes are more delay-intensive on a per-flight basis than MCO's, while MCO's busiest paths carry higher volumes but relatively lower delay per flight, indicating stronger congestion and complexity effects at MIA's core routes.



Path	Direction of Path	Flights	Σ	Total Delay	Total Delay / Flights
ATL-MIA	Incoming	30,063		988,063	33
MIA-ATL	Outgoing	30,058		1,092,426	36
ORD-MIA	Incoming	15,819		835,324	53
MIA-ORD	Outgoing	15,817		877,852	56
MIA-LGA	Outgoing	14,778		798,352	54

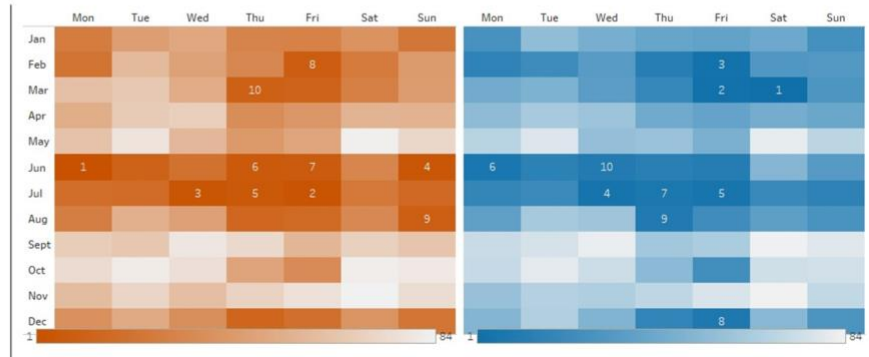
Path	Direction of Path	Flights	Σ	Total Delay	Total Delay / Flights
MCO-ATL	Outgoing	37,711		1,223,217	32
ATL-MCO	Incoming	37,609		1,356,744	36
MCO-PHL	Outgoing	28,357		1,176,659	41
PHL-MCO	Incoming	28,242		1,288,638	46
JFK-MCO	Incoming	23,499		897,450	38

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Seasonality delay analysis shows that both airports are affected by seasonal stress with MCO also having extension to winter and spring.

Both airports share common delay pattern regarding the summer months which indicates shared seasonal stress period. Although MIA shows most high ranked delays in June and July, MCO shows extension to winter and spring, indicating a buildup of congestion rather than a sharper peak at summer.



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Early departure hours show biggest delays while later in the day delays are generally evenly distributed with minor buildups.

MIA and MCO both show pronounced total-delay spikes in the 0-2 departure-hour window; beyond this period, MIA's delays are relatively evenly distributed, while MCO exhibits a modest buildup in delays leading into the early-hour peak.

Departure time radial plots indicating avg total delay using bar height.



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Overall Outcome

Miami International Airport (MIA) demonstrates higher on-time consistency over time but exhibits greater delay variability, higher tail risk, and stronger sensitivity to network and external constraints.

Orlando International Airport (MCO), despite operating at nearly double the traffic volume, shows more predictable delay behavior, lower severity risk, and more stable performance across routes and hours of the day.

As a result, MIA can be characterized as schedule-reliable but risk-prone, while MCO is less dominant in on-time leadership but more operationally stable and resilient.

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Miami International Airport (MIA) - Next Steps

Primary Improvement Goal: Reduce risk and delay volatility while maintaining schedule consistency.

MIA performs well on average punctuality metrics but exposes passengers and operations to greater uncertainty and extreme delay outcomes. Improvement should therefore focus on risk containment, not marginal on-time gains.

Key Improvement Areas:

- * Reduce Delay Variability and Tail Risk.
- * Prioritize improvements that narrow the distribution of delays.
- * Strengthen resilience to NAS.
- * Improve buffering and recovery strategies during high-stress network periods.
- * Focus interventions on the busiest routes where per-flight delays are highest.
- * Reduce external dependency exposure

Expected Outcome:

More predictable passenger experience and reduced exposure to severe delay events without sacrificing on-time pr..

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Orlando International Airport (MCO) - Next Steps

Primary Improvement Goal: Prevent internal delay propagation while preserving operational stability at scale.

MCO demonstrates strong predictability and low cancellation rates despite high traffic volumes. Its main vulnerability lies in delay snowballing, particularly from aircraft rotations and early-day operations.

Key Improvement Areas:

- * Maintain Cancellation Discipline as Volume Grows.
- * Contain Delay Propagation from Late Aircraft.
- * Reduce carry-over effects from early disruptions.
- * Stabilize Early-Day Operations.
- * Address the early-hour delay buildup that drives downstream congestion.

Expected Outcome:

Sustained operational stability and predictability as traffic demand continues to scale.

Disclaimer

Throughout the course of the assignment, all group members contributed equally. While the tasks were divided among us as shown below, we maintained a collaborative approach by actively supporting one another. Whenever difficulties arose or clarification was needed, we engaged collectively to ensure mutual understanding and a cohesive final outcome.

Section	Main Responsible
1	Kitsakis Georgios
2	Schoinas Vasileios
3	Theofanopoulos Michael
4	El Moazen Rami